Optimal-stopping recourse for vehicle routing with simultaneous pickup, delivery, and stochastic demands

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Abstract

In simultaneous pickup-and-delivery operations the load of a vehicle evolves stochastically along its route, and a mid-route capacity breach forces costly emergency recourse. Fleets hold two cheaper levers—handing the remaining route to a standby vehicle, or detouring to the depot to unload collected pickups—whose value depends on the vehicle's position and on prices built from the route's real geometry. We cast the choice of when to exercise which lever as a finite-horizon optimal-stopping problem and solve it with Baton, which adapts regression-based backward induction to this setting: it estimates per-stop continuation costs from historical demand scenarios with a monotone regression, a shape we prove the true continuation value satisfies, and intervenes when continuing becomes dearer than the cheapest recourse action. We show that endpoint-based risk labels are blind to mid-route peaks, that the myopic break-even threshold over-triggers relative to the optimal rule, and that the price of this over-triggering is bounded by the cost of interventions on days that would have completed cleanly and of acting before cheaper later opportunities. On the Dethloff benchmark under six planning regimes, Baton or its state-conditional variant has the

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lowest expected cost of the implementable policies compared—rule-based, position-dependent and two-lever thresholds, depot-return restocking, roll-out and equal-data dynamic programming; a reinforcement-learning policy, compared on a separate set of routes, also trails it. Its margin is 6.5–10.3 percentage points on the five conservative regimes, while on deterministic plans it shows no statistically detectable difference from an equal-data three-action dynamic program. It reaches 86%–96% of the saving of a high-data plug-in program on these regimes and, under independent demand, 94% of the exact optimum on deterministic plans (59% on slack ones); across all benchmarks and demand laws with a non-negligible reactive cost the share ranges from 50% to 100%, lowest where breaches are rare. On the Salhi–Nagy benchmark and on new instances built on the road networks and retail locations of five cities it leads the strongest competitor or shows no statistically detectable difference from it; on the city instances the depot return rarely pays and the policy reduces to its handoff-only form. Because the depot return enlarges the action set, the policy can exceed the clairvoyant bound of the handoff-only class. The conclusions largely survive dependent demand, known day types and a capacitated standby pool shared across routes. The policy has no continuous tuning parameter and fits in milliseconds per route.

Keywords: Routing, stochastic vehicle routing, optimal stopping, recourse policies, last-mile logistics

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1. Introduction

Every morning, an urban parcel operator loads a fleet of vehicles with the day's deliveries and sends each one along a planned sequence of stops. In the simultaneous pickup-and-delivery mode that dominates e-commerce logistics in dense Asian cities—and increasingly elsewhere—each stop is both a delivery and a collection: the driver hands over parcels and takes returns, exchanges, or outbound shipments on board. This operational convenience has an awkward physical consequence. The load of the vehicle no longer

decreases monotonically along the route, as it would in a pure delivery tour; it drifts downward with deliveries and jumps upward with pickups, and on an unlucky day the *running peak* of this load exceeds the vehicle's capacity somewhere in the middle of the tour, even though the final load—everything collected minus everything delivered—would have fit comfortably. When that happens, the operational damage is immediate and expensive. The vehicle can serve no further stop until it is relieved; an ad-hoc replacement must be hired at surge prices to complete the route; and every remaining customer is served late, incurring service-level compensation and goodwill loss. Operators therefore watch their vehicles' loads in real time and, when a breach looks imminent, exercise one of two cheaper forms of recourse. They can *hand off* the remainder of the route to a standby vehicle from a pooled reserve, paying a callout, a per-kilometre surcharge on the remaining tour, and a cargo-transfer dwell. Or, when the depot is near, they can order a *depot return*: the vehicle swings by the depot, unloads the pickups collected so far, and continues its own route with the deliveries still on board, delaying its downstream customers by the detour.

The decision problem this creates is deceptively simple to state and surprisingly subtle to solve. After every stop, given what the vehicle has observed of today's demands so far, should it continue, hand off, or return to the depot? Acting too early wastes money on recourse that hindsight shows was unnecessary—and standby capacity is not free, since every handoff consumes a vehicle-day from the reserve pool. Acting too late means paying the full price of a breach. The trade-off shifts along the route, because both recourse prices fall as fewer stops remain while the scope for demand surprises shrinks; it shifts with the geometry, because a depot return is cheap near the depot and dear far from it; and it shifts with the day, because the demands observed at the first stops carry information about the demands still to come. A policy that captures all three effects must, at every stop, weigh the *cost of continuing optimally*—an expectation over everything that

might still happen, valued under the best future use of both levers—against the current prices of acting now.

The recourse literature for stochastic vehicle routing has approached this class of decisions almost exclusively through *threshold rules*. In the vehicle routing problem with stochastic demands (VRPSD), the classical preventive-restocking policies return the vehicle to the depot when its residual capacity falls below a threshold—a fixed fraction of capacity, a multiple of the next customer's expected demand, or a multiple of the expected demand remaining on the route (Yang et al., 2000; Salavati-Khoshghalb et al., 2019). Threshold rules are attractive because they are transparent and easy to implement, and with carefully tuned parameters they capture a useful share of the available saving. But they hard-wire two approximations that our setting exposes. First, a threshold on the current load ignores *option value*: early in the route, waiting is nearly free because many opportunities to act remain, while late in the route the same load level may leave a single chance to intervene. A fixed threshold cannot be right at both ends, and we prove below that the myopic, break-even threshold errs in one direction: it over-triggers. Second, and more fundamentally, when recourse prices vary by stop—as they do whenever they are built from real route geometry—the optimal intervention boundary is not representable by *any* single threshold on load or on breach probability: the comparison that defines optimal behaviour is between two costs, not between a probability and a constant.

This paper develops, analyses, and benchmarks a policy that makes exactly that comparison. Baton (*Backward-induction AcTion pricing for ONline recourse*) learns, from nothing more than historical demand realisations, a per-stop estimate $\widehat{C}_k(w)$ of the expected cost of continuing past stop k with load state w under optimal future play, by sweeping the historical paths backwards in the manner of Longstaff and Schwartz (2001) and projecting each regression onto the class of monotone functions—a shape restriction we prove the true continuation value satisfies when the load evolves as a stochastically

monotone Markov chain. Online, the vehicle performs a single comparison after each stop: it exercises the cheapest recourse action whose price is below the estimated cost of continuing, and otherwise drives on. The policy has no continuous tuning parameter (its one discrete choice, the recourse menu, is made on the training days); the threshold rules of the literature are recovered as its myopic degeneration. The same backward induction prices a menu of recourse actions: adding the depot return requires one extra term in the Bellman recursion—the continuation value from the post-return state, which the algorithm evaluates by simulating the reset state through its own already-fitted downstream policy—and the resulting three-action policy chooses, stop by stop and day by day, whichever lever the economics favour.

The paper makes three contributions, and we separate them because they are of different kinds. The first is a *model*: execution-stage recourse in simultaneous pickup-and-delivery routing, with its non-monotone load, formulated as a finite-horizon optimal-stopping problem over a menu of levers whose prices are built per stop from the route's geometry and a three-class fleet economy. The second is a set of *structural results* about that model: endpoint-based risk labels are blind to mid-route peaks, and we bound the excess cost this causes (Proposition 1); the stopping region of the myopic (break-even) rule contains the optimal one, so the break-even threshold over-triggers (Proposition 2), and the price of this over-triggering is bounded by the cost of interventions on days that would have completed cleanly, plus any saving forgone by acting before a cheaper later opportunity (Proposition 3); and the continuation value is monotone in the load (Proposition 4), which justifies a nonparametric monotone estimator. All four propositions are machine-checked in the Lean 4 proof assistant with the Mathlib library, and the formal proofs are released with the paper; the formalisation exposed a side condition of Proposition 1 (at least one decision epoch before the last stop) that the informal statement had left implicit. The third is a *method* that is deliberately an adaptation rather than an invention: regression-based

backward induction for optimal stopping (Longstaff and Schwartz, 2001; Tsitsiklis and Van Roy, 2001; Clément et al., 2002) is established, and we claim novelty only for the pieces the model requires—the isotonic regression step licensed by Proposition 4, the simulation-based valuation of the post-return state, and the train-set selection of the recourse menu.

We evaluate the policy in an extensive computational study resting on three pillars. The first is breadth of comparison: we evaluate against the published rule-based restocking family of Salavati-Khoshghalb et al. (2019), a depot-return restocking policy in the tradition of Florio et al. (2020) and Legault et al. (2025), a rollout policy in the spirit of Secomandi (2001) and a cost-scaled version of it, a tuned threshold on our corrected risk labels and a position-dependent threshold with one tuned level per stop, a reinforcement-learning policy re-implemented from Iklassov et al. (2024), a two-lever rule that gives a tuned threshold the depot return as well, and dynamic-programming benchmarks—two restricted to the same data budget as BATON and two supplied with fifty times more data as high-data references. The second pillar is breadth of environment: six planning regimes generate the routes, from a deterministic gate through sample-average and Wasserstein distributionally robust gates to three published robust planners (Gounaris et al., 2013; Bertsimas and Sim, 2004; Ghosal et al., 2024); the instances span the classical Dethloff and Salhi–Nagy benchmarks and a new set built on the actual road networks of Ho Chi Minh City, Hanoi, New York, Paris, and Shanghai, with customers placed at real shop locations. The third pillar is anchoring: the execution stage is compared against an exact dynamic program where one can be computed (independent demand), against high-data plug-in programs elsewhere, and against clairvoyant bounds for both action menus; the planning stage is certified against mixed-integer programming bounds computed with a commercial solver. The study also probes the assumptions behind the model—demand dependence, known day types, the post-return state, a capacitated standby pool shared across routes, and the

size of the training history.

On the Dethloff benchmark, Baton or its state-conditional variant has the lowest expected execution cost of the implementable policies on every planning gate. On the five conservative gates its margin over the strongest competitor is 6.5–10.3 percentage points with confidence intervals that exclude zero; on the deterministic gate it shows no statistically detectable difference from an equal-data three-action dynamic program. It reaches 86%–96% of the saving of a high-data plug-in program for its own action set on these gates, and 94% of the exact optimum on the deterministic plans where the exact optimum can be computed (59% on the slack SAA plans). Across all our benchmarks and demand laws with a reactive cost of at least three units per plan-day that share ranges from 50% to 100%; it is lowest where breaches are rare, because a thousand days then carry little information about them. On four of the six gates Baton exceeds the clairvoyant bound of the handoff-only class. This is not a paradox: that bound is clairvoyant only about the handoff lever, and a depot return can prevent a breach at a price no handoff can match, so the excess measures the value of the second lever; the clairvoyant bound for the full menu stays well above every policy. A two-lever threshold rule that also has the depot return captures part of this value but stays 1.4–10.3 points below Baton, so the gain comes from pricing the levers, not merely from having them. The economics sweeps deliver the managerial headline: when standby capacity is expensive, threshold policies lose most of their saving while Baton shifts its recourse mix toward depot returns; when service-level penalties rise, it shifts back—without re-tuning, because the policy prices actions rather than memorising trigger levels. The gains are not uniform, and we report where they are small or absent. On the city instances the depot return rarely pays and Baton reduces to its handoff-only form, leading the best threshold by only +0.5 points; with half of the customers deliver-only it shows no statistically detectable difference from an equal-data dynamic program; and on tightly packed routes a well-started

position-dependent threshold comes close to the handoff-only policy.

The remainder of the paper unfolds as follows. Section 2 positions the work within the recourse, robust-planning, and learning literatures. Section 3 develops the operational setting and cost model, formulates the stopping problem, diagnoses why endpoint labels and fixed thresholds fail, and presents the BATON policy with its supporting analysis and algorithms. Section 4 reports the computational study. Section 5 distils the managerial insights, states the scope of the conclusions and their limitations, and concludes.

2. Related work

The vehicle routing problem with stochastic demands is one of the oldest and most studied stochastic combinatorial problems in operations research; the surveys of Gendreau et al. (1996) and Oyola et al. (2018) trace its development, and Koç et al. (2020) review the simultaneous pickup-and-delivery variant that concerns us. Within this literature our work sits at the intersection of three strands: recourse policies for the execution stage, robust and distributionally robust planning for the construction stage, and learning-based approaches that estimate policies from data.

The recourse strand begins with the a priori optimisation paradigm of Bertsimas (1992) and Dror et al. (1989), in which routes are fixed before demands are revealed and a simple rule—return to the depot upon failure—prices the risk. The insight that *preventive* action dominates reactive failure recovery led to the restocking policies of Yang et al. (2000), who compute optimal return thresholds by dynamic programming on a fixed route and show that the optimal policy has one threshold per stop. For simultaneous pickup and delivery on a predefined customer sequence, Minis and Tatarakis (2011) derive the corresponding dynamic program and threshold-type optimal restocking policies, and Tatarakis and Minis (2009) allow multiple depot returns; our exact benchmark in §4.5 solves the same kind of fixed-sequence

program for the handoff and return levers. These results rest on the classical theory of monotone optimal policies for Markov decision processes (Serfozo, 1976; Puterman, 1994) and on stochastic-order comparison arguments (Müller and Stoyan, 2002), of which our Proposition 4 is an instance. They led also to a rich family of rule-based triggers whose modern form is analysed by Salavati-Khoshghalb et al. (2019): return when residual capacity falls below a fixed fraction of capacity, below a multiple of the next customer's expected demand, or below a multiple of the expected demand remaining. Exact algorithms that price optimal restocking within route optimisation culminate in Florio et al. (2020) and, most recently, in superadditivity-based formulations (Legault et al., 2025) and scenario-optimal recourse under empirical demand distributions (Ota and Fukasawa, 2026)—the latter sharing our data-driven stance, though at the planning rather than the execution stage. Reoptimisation approaches solve the execution stage as a Markov decision process: Secomandi and Margot (2009) characterise partial-reoptimisation policies, and the rollout algorithm of Secomandi (2001) remains the standard implementable approximation, improving a base policy by one step of policy iteration. Our contribution to this strand is a policy that retains the implementability of the rule-based family while provably escaping its two structural approximations—the endpoint-for-peak label substitution and the threshold functional form—and that generalises across recourse menus, pricing the depot return of the restocking tradition and the vehicle handoff of platform logistics within one recursion.

The robust-planning strand constructs routes that remain feasible when demands deviate. Gounaris et al. (2013) formulate the robust capacitated VRP under polyhedral demand uncertainty and derive the quadrant-budget sets we use as one of our planning gates; Gounaris et al. (2016) give an adaptive memory metaheuristic, and Pessoa et al. (2021) the state-of-the-art exact algorithm, for the same problem class. Budget uncertainty in the sense of Bertsimas and Sim (2004) supplies a second gate, and the unifying distribu-

tionally robust framework of Ghosal et al. (2024), whose moment-ambiguity feasibility check reduces to a Cantelli-type closed form, a third. Sample-average and Wasserstein-ambiguity gates (Mohajerin Esfahani and Kuhn, 2018) complete the planning spectrum. We treat these planners as *environments* rather than competitors: execution recourse operates on whatever routes planning produces, and one of our findings is that planning conservatism and execution intelligence are substitutes at the margin—conservative plans leave little for any execution policy to save, yet it is precisely there that pricing the full action menu matters most, because on such plans the threshold family saves little or nothing while the priced policy still extracts double-digit savings.

The learning strand estimates routing policies from data. Approximate-dynamic-programming approaches with lookup tables (Novoa and Storer, 2009) and offline value-function approximation (Ulmer et al., 2019) established that learned continuation values can guide online routing decisions; the recent wave of neural constructive methods extends this to end-to-end route construction, with Iklassov et al. (2024) the representative treatment of the stochastic demand case and Kadyrov et al. (2025) a recent application to dynamic settings; Hu and Tao (2026) and Wang et al. (2026) study uncertainty in simultaneous pickup-and-delivery variants adjacent to ours. Our method is deliberately conservative within this strand: it uses the regression-based backward induction of American-option pricing (Longstaff and Schwartz, 2001; Tsitsiklis and Van Roy, 2001), whose convergence with fixed and with nonparametric approximation classes has been analysed (Clément et al., 2002; Egloff, 2005; Zanger, 2013), combined with isotonic regression (Barlow et al., 1972; Ayer et al., 1955) whose shape constraint we justify structurally. Two of our empirical results speak directly to this strand: a policy-gradient network trained on the same data captures materially less of the achievable saving than the monotone regression, and enriching the conditioning statistic with factor posteriors and recency features—the natural representation-

learning move—makes matters slightly worse at operational sample sizes. We read these results as statements about a regime, not about learning in general: with a scalar sufficient state, a known monotone value structure and a thousand paths per route, a shaped regression is hard to beat, and §4.4 discusses which learning architectures could narrow the gap as the state or the data grow.

3. The execution problem and the BATON policy

This section develops the paper's methodological content as one continuous argument: the operational setting and its cost structure (§3.1), the optimal-stopping formulation (§3.2), the two structural defects of the incumbent approach and their formal diagnosis (§3.3), the estimation scheme (§3.4), the generalisation to the full recourse action set (§3.5), and the resulting algorithms with their complexity and reference bounds (§3.6). The four propositions have been verified formally in the Lean 4 proof assistant with the Mathlib library; the formal statements, and how they correspond to the text, are part of the replication repository (see Data availability). The formalisation abstracts from the text in three places: the transition law is an abstract kernel carrying only the stochastic-monotonicity property of Assumption 1; the inclusion and strictness claims of Proposition 2 and Proposition 3 are proved on finite history trees, so they cover rules that may use the whole observed history rather than the deployed rules, which observe W_k only; and the class of admissible continuations is left abstract, with only the properties the proofs use. On finite-support demand distributions—including the empirical distribution of the training days—the tree versions of Propositions 2 and 3 hold without the Markov Assumption 1, with continuation values conditioned on the history rather than on W_k alone; they do not by themselves establish the claims for the W_k-rules that BATON deploys, which rest on Assumption 1.

3.1. Operational setting and the three-class fleet cost model

Consider one planned route serving m customers in a fixed sequence $1, \ldots, m$ from a depot. The vehicle departs carrying the route's deliveries; at stop k it delivers a random volume $d_k \geq 0$ and collects a random pickup $p_k \geq 0$, each observed on service, producing the *net increment*

$$g_k \;=\; p_k - d_k. \tag{1}$$

The *running net load* is

$$W_k \;=\; \sum_{j=1}^{k} g_j, \qquad W_0 = 0, \tag{2}$$

and with departure load L_0 and capacity Q the vehicle physically overflows at stop k when $L_0 + W_k > Q$, that is, when W_k exceeds the *slack* $B = Q - L_0$. The *breach time* is

$$T \;=\; \min\{\, k \leq m : W_k > B \,\}, \qquad \min \emptyset := \infty, \tag{3}$$

and the route *peak-overflows* on the event $\{T \leq m\} = \{\max_{k \leq m} W_k > B\}$. We write $\mathcal{F}_k = \sigma(g_1, \ldots, g_k)$ for the information available after serving stop k.

Loading is *nominal*: the vehicle departs with the planned delivery volume $L_0 = \sum_k \bar{d}_k$, and the realised delivery d_k is revealed on service. The capacity side of the problem, the only one that triggers recourse, is then $L_0 + W_k \leq Q$. A realised delivery can exceed the cargo still on board; in our benchmark protocol the day's deliveries exceed L_0 on 47% of route-days, by 10% of L_0 on average. Such shortfalls are outside the model: they are not priced and no policy affects them, so they do not change the comparison of policies. The alternative convention, loading the realised delivery volume, would make L_0 random but known at departure and would reveal the day's delivery total before the first stop; it changes the information structure and is not studied

here. A route that departs fully loaded has $B = 0$ and breaches as soon as its running load rises; 16 routes of the large-scale benchmarks of §4.3 (none of the Dethloff routes) depart this way.

Costs follow the economics of an urban last-mile operator running three vehicle classes. The *planned* fleet is contracted at a day rate F_{plan} per vehicle plus a running cost c_{km} per kilometre; these are sunk once the plan is fixed and play no role in execution decisions. The *standby* class supplies planned handoffs: capacity is pooled across the operation, and a handoff after stop k consumes one standby vehicle-day at rate F_{sb}, a dispatch fee F_{ho} (the fixed charge for calling out a standby vehicle), a per-kilometre surcharge s_{ho} on the remaining tour δ_k (the distance of legs $k \to k{+}1 \to \ldots \to m \to$depot), and a cargo-transfer dwell c_{tr}:

$$H_k \;=\; F_{\text{sb}} + F_{\text{ho}} + s_{\text{ho}}\, c_{\text{km}}\, \delta_k + c_{\text{tr}}. \tag{4}$$

The *emergency* class is hired ad hoc at surge prices when a breach actually occurs at stop k: a surge callout F_{emg}, the remaining distance at a surge multiplier s_{emg}, a service-level compensation p_{late} for each of the $m-k$ downstream customers served late, and a goodwill loss at the breached stop itself,

$$E_k \;=\; F_{\text{emg}} + s_{\text{emg}}\, c_{\text{km}}\, \delta_k + p_{\text{late}}\,(m-k) + p_{\text{breach}}. \tag{5}$$

Finally, a *depot return* after stop k sends the vehicle from stop k to the depot and back to stop $k{+}1$, unloading the pickups collected so far while the undelivered cargo stays on board; it costs the detour at the running rate plus the transfer dwell plus the same per-customer lateness as any other delay,

$$R_k \;=\; c_{\text{km}}\big(D_{k,0} + D_{0,k+1} - D_{k,k+1}\big) + c_{\text{tr}} + p_{\text{late}}\,(m-k), \tag{6}$$

where $D_{i,j}$ denotes road distance. All three schedules are computed from the route's actual geometry before departure and are therefore known constants

at decision time; both H_k and E_k decrease along the route. At our default parameters

$$E_k \;>\; H_k \quad \text{for all } k, \tag{7}$$

an unplanned breach at a stop being dearer than a planned transfer at the same stop. Inequality (7) describes the calibration, not a requirement of the method: none of our results uses it (Assumption 2 below asks only that E_k be non-increasing), and where it fails—when standby capacity is priced above an emergency hire, as in one configuration of §4.8—the handoff is simply dominated at the stops concerned and the policy never uses it there. The flat two-price model $H_k \equiv \omega_F$, $E_k \equiv C_{\text{fail}}$ with $C_{\text{fail}} > \omega_F$ is the special case used in our controlled experiments and by the incumbent system.

Standby versus emergency vehicles. The two recourse classes differ in how they are contracted and in what they cost the customer, not in the vehicles themselves. A *standby* vehicle is reserved capacity under a pre-agreed rate card: it is dispatched *before* a breach, meets the planned vehicle at the next stop, takes over the remaining route at a small per-kilometre surcharge, and keeps the downstream customers on schedule apart from the transfer dwell. An *emergency* vehicle is hired on the spot market *after* the planned vehicle has overflowed: it is paid a surge callout and surge distance rate, must first reach a vehicle that is already stranded, and every downstream customer is served late. The standby day rate F_{sb} in (4) is the holding cost of standby capacity, charged per vehicle-day actually consumed. This pay-per-use convention treats the standby pool as shared across a metropolitan operation and large relative to one plan, so that a single plan's handoffs never exhaust it; it is the standard arrangement with third-party reserve fleets, and it makes the pool size a by-product of the policy (the 95th percentile of daily handoffs, reported with our results) rather than a constraint on it. When instead a fixed number of standby vehicles is reserved in advance, their holding cost is sunk at execution time and the pool is a shared, capacitated

resource that couples the routes; §4.6 treats that case explicitly.

3.2. The optimal-stopping formulation

Restrict attention for the moment to the handoff lever alone; the general action set is treated in §3.5. An *execution policy* is then a stopping time σ with respect to $(\mathcal{F}_k)$ taking values in $\{1, \dots, m-1\} \cup \{\infty\}$, with $\sigma = \infty$ meaning never hand off, and the realised execution cost of a day is

$$J(\sigma) \;=\; \begin{cases} E_T, & T \le \sigma \wedge m \quad \text{(breach first)}, \\ H_\sigma, & \sigma < T \quad \text{(handoff first)}, \\ 0, & \sigma = \infty,\ T = \infty \quad \text{(clean completion)}. \end{cases} \tag{8}$$

The operator seeks $\inf_\sigma \mathbb{E}\, J(\sigma)$, the expectation taken over the demand law. Throughout the analysis we assume:

Assumption 1. The running load $(W_k)_{k=0}^m$ is a Markov chain with respect to $(\mathcal{F}_k)$, possibly time-inhomogeneous, whose transition kernels are *stochastically monotone*: for every k and all $x \le x'$, the conditional law of W_{k+1} given $W_k = x$ is dominated in the usual stochastic order by its conditional law given $W_k = x'$.

Assumption 2. The emergency schedule (E_k) is non-negative and non-increasing in k; the handoff and return schedules (H_k), (R_k) are arbitrary non-negative constants.

Assumption 1 makes (k, W_k) a sufficient state for the handoff problem and covers two cases of practical interest. The first is independent increments, not necessarily identically distributed: the conditional law of W_{k+1} given $W_k = x$ is that of $x + g_{k+1}$, which is stochastically increasing in x. The second is positively dependent demand driven by a common day-level factor: if $g_j = \mu_j + \beta Z + \varepsilon_j$ with a Gaussian factor Z and independent Gaussian noise ε_j of common variance, the likelihood of $(g_1, \dots, g_k)$ depends on Z only through

W_k, so W_k is sufficient for the factor, the posterior of Z is increasing in W_k, and the chain is Markov with stochastically monotone kernels. Outside these cases W_k is not exactly sufficient—in our benchmark protocol pickups and deliveries load on separate factors with customer-specific scales—and the policies of this paper are then optimal within the class of policies that observe (k, W_k) only in the approximate sense of regression-based stopping. Section 4.5 measures what this costs: it varies the demand correlation from 0 to 0.9, adds a multiplicative day factor, and tests the monotonicity of the estimated continuation value directly. Assumption 2 holds by construction of (5), since both the remaining distance and the number of downstream customers fall along the route. By finite-horizon dynamic programming, define value functions backwards from $V_m(w) \equiv 0$ on $\{w \le B\}$ by

$$C_k(w) \;=\; \mathbb{E}[\, E_{k+1}\, \mathbf{1}\{W_{k+1} > B\} \;+\; V_{k+1}(W_{k+1})\, \mathbf{1}\{W_{k+1} \le B\} \,|\, W_k = w\,], \tag{9}$$

$$V_k(w) \;=\; \min\{\, H_k,\; C_k(w)\,\}, \qquad k = m-1, \ldots, 1, \tag{10}$$

where $C_k(w)$ is the expected cost of *continuing past stop k and playing optimally thereafter*. The optimal policy is the first-passage rule

$$\sigma^\star \;=\; \min\{\, k < m \wedge T \;:\; C_k(W_k) > H_k \,\} \tag{11}$$

by the standard verification argument for finite-horizon stopping problems (Chow et al., 1971; Peskir and Shiryaev, 2006). Everything difficult about the problem is therefore statistical: the demand law is unknown, so C_k must be estimated from historical load paths, and the quality of the policy is the quality of that estimate in the region where the comparison (11) is close.

3.3. Two structural defects of the incumbent approach

The system this work replaces—like much of the rule-based literature—estimated a per-stop breach *probability* $p_k(w)$ by isotonic regression of a

binary label on W_k across historical routes and triggered a handoff when $\hat{p}_k(W_k) > \tau$ for a tuned constant τ. Both design choices fail structurally, not statistically. Both failures were first noticed in the operation that motivated this work, before they were understood formally; these observations are anecdotal, and the evidence we offer for both defects is the analysis below and the controlled experiments of §4.

The first defect concerns the label. The natural per-route summary in logged data is the endpoint total W_m, and the incumbent labelled a route as overflowing by $Y^{\text{end}} = \mathbf{1}\{W_m > B\}$ rather than by the physical event $Y^{\text{peak}} = \mathbf{1}\{\max_{k \le m} W_k > B\}$. We state the consequence formally; despite its simplicity, it explains an observed (anecdotal) failure in which a deployed policy saved nothing, and the same endpoint-for-peak substitution recurs whenever monitoring data are aggregated per route.

Proposition 1 (Endpoint-label bias)**.** *For every path law and every B, $\{W_m > B\} \subseteq \{\max_{k \le m} W_k > B\}$, hence $\mathbb{P}(Y^{\text{end}}{=}1) \le \mathbb{P}(Y^{\text{peak}}{=}1)$, and the endpoint-trained regression estimates a conditional probability of the wrong event. The gap is extremal on collect-then-deliver routes: if the route structure forces $W_m \le 0$ almost surely and $B > 0$, then $Y^{\text{end}} = 0$ a.s. while $\mathbb{P}(T \le m)$ may be arbitrary in $[0, 1]$. In that regime every threshold policy trained on endpoint labels coincides with the reactive policy and forgoes the entire value of preventive recourse, $\Delta := V^{\text{react}} - V^{\star}$, where $V^{\text{react}} = \mathbb{E}[E_T \mathbf{1}\{T \le m\}]$ and $V^{\star} = \inf_{\sigma} \mathbb{E}J(\sigma)$. Under flat prices $(\omega_F, C_{\text{fail}})$ with $0 \le \omega_F \le C_{\text{fail}}$ and routes with $m \ge 2$ stops (so that at least one decision epoch precedes the last stop),*

$$\max\big\{0,\ (C_{\text{fail}} - \omega_F)\,\mathbb{P}(2 \le T \le m) - \omega_F\,\mathbb{P}(T = \infty)\big\} \le\ \Delta\ \le\ (C_{\text{fail}} - \omega_F)\,\mathbb{P}(2 \le T \le m), \quad (12)$$

and the upper bound is attained when the breach is announced one stop ahead, that is, when $T - 1$ is a stopping time on $\{2 \le T \le m\}$.

The proof, elementary but careful about the decision epochs, is given in

Appendix Appendix A.1.

The bounds (12) say that when breaches are frequent relative to clean days, the endpoint label costs the operator almost the whole value of the handoff lever, $(C_{\text{fail}} - \omega_F)$ per breach that could have been anticipated. The synthetic collect-then-deliver scenario of §4.2 realises this regime: the endpoint-trained policy saves exactly nothing there. BATON therefore labels and calibrates on the running peak everywhere; in particular, when the slack must be calibrated empirically, it uses the $(1-\alpha)$-quantile of historical *peaks*, never of endpoint totals.

The second defect concerns the trigger. Even with the label corrected, a myopic comparison is the wrong functional form, and its error has a sign. Write $C_k^0(w) = \mathbb{E}[E_T \mathbf{1}\{T \le m\} \mid W_k = w, T > k]$ for the cost of continuing from (k, w) and never acting again. The *myopic rule* hands off after stop $k < m \wedge T$ when $C_k^0(W_k) > H_k$. Under flat prices $C_k^0(w) = C_{\text{fail}}\, p_k(w)$ with $p_k(w) = \mathbb{P}(T \le m \mid W_k = w, T > k)$ the peak-overflow probability, and the myopic rule is exactly the probability threshold $p_k(w) > \tau$ with $\tau = \omega_F / C_{\text{fail}}$.

Proposition 2 (Myopic over-triggering)**.** *Under Assumption 1, at every stop $k < T$ the stopping region of the optimal rule (11) is contained in that of the myopic rule:*

$$\{\, w : C_k(w) > H_k \,\} \;\subseteq\; \{\, w : C_k^0(w) > H_k \,\}.$$

Moreover $C_k(w) < C_k^0(w)$ whenever, with positive probability given $W_k = w$ and $T > k$, the path survives to some stop $j \in \{k+1, \dots, m-1\}$ at which the myopic rule itself fires. If, in addition, Assumption 2 holds, C_k and C_k^0 are continuous, and this condition holds at the myopic boundary $\bar{w}_k = \inf\{w : C_k^0(w) > H_k\} < B$, the inclusion is strict: a right-neighbourhood of $\bar{w}_k$ is stopped by the myopic rule but continued by the optimal one.

Proof. "Continue and never act again" is an admissible continuation from (k, w) with expected cost $C_k^0(w)$; since $C_k(w)$ is the infimum over admissible

continuations, $C_k \le C_k^0$, which gives the inclusion. For the strict inequality, let ρ be the first stop $j > k$ with $T > j$ and $C_j^0(W_j) > H_j$, and consider the continuation that hands off at ρ and never acts otherwise. It differs from "never act" only on $A = \{\rho < m\}$, where it pays H_ρ while never acting costs, conditionally on $\mathcal{F}_\rho$ and by the strong Markov property, $C_\rho^0(W_\rho) > H_\rho$. If $\mathbb{P}(A \mid W_k = w, T > k) > 0$, its expected cost is strictly below $C_k^0(w)$, and so is $C_k(w)$. Finally, C_k^0 is non-decreasing (it is the continuation value of the problem with $H \equiv \infty$, to which Proposition 4 applies), so $C_k^0 > H_k$ on $(\bar{w}_k, B]$ and, by continuity, $C_k^0(\bar{w}_k) = H_k$. Then $C_k(\bar{w}_k) < C_k^0(\bar{w}_k) = H_k$, and continuity of C_k gives $C_k < H_k < C_k^0$ on a right-neighbourhood of $\bar{w}_k$. □

Proposition 2 is qualitative. The next result prices the over-triggering. Let σ^0 and $\sigma^\star$ be the stopping times of the myopic and the optimal rule, and write $L_k := 0$ on $\{T = \infty\}$ and $L_k := \min\{\min_{k<j<T} H_j,\ E_T\}$ on $\{k < T \le m\}$ for the clairvoyant cost of the remainder of the day after stop k: the cheapest later handoff before the breach, or the breach itself.

Proposition 3 (The price of over-triggering)**.** *Under Assumption 1, $\sigma^0 \le \sigma^\star$ almost surely, and*

$$\begin{aligned} 0 \;\le\; \mathbb{E}\,J(\sigma^0) - \mathbb{E}\,J(\sigma^\star) \;&=\; \mathbb{E}\big[\big(H_{\sigma^0} - C_{\sigma^0}(W_{\sigma^0})\big)\mathbf{1}\{\sigma^0 < \sigma^\star\}\big] \\ &\le\; \mathbb{E}\big[\big(H_{\sigma^0} - L_{\sigma^0}\big)\mathbf{1}\{\sigma^0 < \sigma^\star\}\big]. \end{aligned} \tag{13}$$

Under flat prices the right-hand side is at most $\omega_F\,\mathbb{P}(\sigma^0 < \sigma^\star, T = \infty) \le \omega_F\,\mathbb{P}(\sigma^0 < \infty, T = \infty)$.

Proof. If the optimal rule stops at k then W_k lies in its stopping region, hence in the myopic one by Proposition 2; the myopic rule has therefore stopped at k or earlier, so $\sigma^0 \le \sigma^\star$. On $\{\sigma^0 = \sigma^\star\}$ the two policies take identical actions all day and pay the same. On $\{\sigma^0 < \sigma^\star\}$, an event in $\mathcal{F}_{\sigma^0}$, the myopic rule pays H_{σ^0} while the optimal rule continues from (σ^0, W_{σ^0}), whose conditional expected cost is $C_{\sigma^0}(W_{\sigma^0})$ by the strong Markov property. This gives the

identity, and the difference is non-negative because $\sigma^\star$ is optimal. For the inequality, every admissible continuation from stop k pays, path by path, either some H_j with $k < j < T$, or E_T, or nothing on a clean day, hence at least L_k; taking conditional expectations, $C_k(W_k) \geq \mathbb{E}[L_k \mid \mathcal{F}_k]$ on $\{T > k\}$, and the tower property gives the bound. Under flat prices $H_{\sigma^0} - L_{\sigma^0}$ equals ω_F on $\{T = \infty\}$, 0 on $\{T \geq \sigma^0 + 2\}$ and $\omega_F - C_{\text{fail}} < 0$ on $\{T = \sigma^0 + 1\}$. □

Proposition 3 turns the critique into an accounting statement. The premium a threshold rule pays for its simplicity is at most the handoff price times the probability that it intervenes on a day that would have ended cleanly, plus, under declining prices, the saving forgone by handing off before a cheaper later opportunity. The bound gives a sufficient condition for the myopic threshold to be "good enough": if both terms are small, so is the loss. This is the case when the states in which it fires lead to a breach almost surely, as on tightly packed routes whose risk sits in the next stop or two, and when prices decline slowly along the route. The condition is not necessary, since the bound need not be tight. The bound is larger on plans that carry slack, where the rule fires on more days that would have recovered by themselves. Section 4.4 evaluates both sides of (13) on our benchmarks.

Proposition 2 is also consistent with an anecdotal observation from the same operation: tuned thresholds drifted upward—toward less triggering—precisely on long routes, chasing option value the functional form cannot express. Under state-dependent prices the failure is also one of representation: the comparison $C_k^0(w) > H_k$ with per-stop H_k is not a single threshold on p_k, whereas the cost comparison of (11) is unchanged. A threshold that varies by stop can represent the optimal boundary—since C_k is monotone (Proposition 4), the optimal stopping region at each stop is an upper set $\{w > w_k^\star\}$—but its $m - 1$ boundaries must then be searched directly against realised cost; §4.1 includes such a position-dependent threshold, and a cost-scaled myopic rule, among the competitors, to isolate the value of *estimating* the boundary by backward induction rather than searching for it.

3.4. Estimating continuation values from historical paths

Estimating (9) requires a conditional expectation of future costs *under optimal future play*—precisely the structure of American-option pricing by simulation, and we adopt the regression-based backward induction of Longstaff and Schwartz (2001). Sweep the N historical paths backwards, maintaining for each path s the realised future cost ϕ_s under the policy already constructed for later stops, initialised by $\phi_s = E_{T_s}$ if path s breaches and 0 otherwise; at stop k, regress ϕ on W_k among the paths still alive (those with $T_s > k$) to obtain $\widehat{C}_k$; then apply the implied action, replacing ϕ_s by H_k on paths where $\widehat{C}_k(W_{k,s}) > H_k$, so that stop $k-1$ sees costs under the improved later policy. The regression target is a cost supported on $\{0\} \cup \{H_j\}_{j>k} \cup \{E_j\}_{j>k}$, not a probability, and the estimator we use for every regression is *isotonic* least squares (Barlow et al., 1972). The shape constraint is not a convenience; it is a theorem about the estimand:

Proposition 4 (Monotone continuation value)**.** *Under Assumptions 1 and 2, for every $k \in \{1, \dots, m-1\}$ the continuation value $C_k(\cdot)$ of (9) is non-decreasing on $(-\infty, B]$. The same holds for any recourse menu whose actions either end the route at a price known at stop k or reset the load to a fixed level at a known price—in particular for the three-action recursion (14) below, and for the menu without the handoff lever, whose continuation value is C_k^0.*

Proof. Backward induction on k. Write the integrand of (9) as $\varphi_{k+1}(x) = E_{k+1}\mathbf{1}\{x > B\} + V_{k+1}(x)\mathbf{1}\{x \le B\}$, so that $C_k(w) = \mathbb{E}[\varphi_{k+1}(W_{k+1}) \mid W_k = w]$. By Assumption 1 the conditional law of W_{k+1} is stochastically increasing in w, so C_k is non-decreasing as soon as φ_{k+1} is. At $k+1 = m$ we have $V_m \equiv 0 \le E_m$, so φ_m is a non-decreasing step function. For $k+1 < m$, the inductive hypothesis makes C_{k+1} non-decreasing, and V_{k+1} is the minimum of C_{k+1} and of constants (the handoff price, and the return price plus the fresh-start value), hence non-decreasing on $(-\infty, B]$. Moreover $V_{k+1}(x) \le C_{k+1}(x) \le E_{k+1}$: the cost of continuing and never acting again is at most the largest emergency price still reachable, $\max_{j \ge k+2} E_j$, which is at most E_{k+1} by

Assumption 2. For $x \le x'$: if both exceed B, $\varphi_{k+1}(x) = \varphi_{k+1}(x') = E_{k+1}$; if both are at most B, $\varphi_{k+1}(x) = V_{k+1}(x) \le V_{k+1}(x') = \varphi_{k+1}(x')$; if $x \le B < x'$, then $\varphi_{k+1}(x) = V_{k+1}(x) \le E_{k+1} = \varphi_{k+1}(x')$. Thus φ_{k+1} is non-decreasing in every case, completing the induction. No ordering between H_k (or R_k) and E_k is used. □

The argument is the classical one for monotone optimal policies (Serfozo, 1976), (Puterman, 1994, §4.7); we include it because the peak-breach terminal cost and the resetting action need the two case distinctions above.

Isotonic regression is therefore the least-squares projection onto exactly the function class known to contain the truth; it requires no basis choice, bandwidth, or regularisation parameter, and it inherits classical consistency guarantees (Ayer et al., 1955; Barlow et al., 1972). For the full backward-induction scheme, the analysis of Clément et al. (2002) covers a fixed, finite linear basis and does not apply to isotonic regression. The statistical-learning analyses of Egloff (2005) and Zanger (2013) treat nonparametric approximation classes of controlled complexity; uniformly bounded monotone functions on an interval have finite bracketing entropy, which suggests that their arguments extend to the isotonic step, but we do not prove a convergence theorem for Baton here. One feature of the scheme matters for reading the shape constraint: the regression target at stop k is the realised cost under the *fitted* downstream policy, not under the optimal one, so it is monotone in W_k only in the limit in which the downstream fits converge; in finite samples the isotonic step projects a target that may be slightly non-monotone. When Assumption 1 holds only approximately, the isotonic fit is the monotone projection of the conditional expectation given W_k; §4.5 checks directly, on fifty thousand paths per route, whether that conditional expectation is monotone under strongly dependent demand.

Remark 1. Backward induction propagates estimation error from late stops into the regression targets of early stops, so its advantage over a directly tuned threshold should shrink as the history shortens. An earlier version of

this analysis therefore recommended falling back to the label-corrected tuned threshold below about 500 training paths. The data-budget experiment of §4.7 does not support that recommendation: BATON beats the tuned threshold at every history length we tested, down to 100 training days per route. The isotonic projection pools information along the load axis, which keeps the propagated error small. The regime in which backward induction does over-fit is a different one—breaches so rare that a history contains only a few dozen of them (§4.5). There the full policy learns to decline acting, while its handoff-only restriction still over-fits and loses a little money.

3.5. Pricing the full action set

The depot-return transition. After stop k the vehicle carries $L_0 + W_k$, where L_0 is the departure load. A return drives to the depot, unloads every pickup collected so far, and sends the vehicle back to stop $k+1$ to continue its route in the planned order, at the price R_k of (6). The policy records the post-return state as $W = 0$ against the unchanged slack B, that is, as if the vehicle left the depot again with its departure load L_0. This is a conservative convention. Unloading the pickups leaves on board exactly the undelivered cargo $\max\{L_0 - D_{\le k}, 0\}$, where $D_{\le k} = \sum_{j \le k} d_j$ is the volume already delivered, so the physical post-return state is $x_k = -\min\{D_{\le k}, L_0\} \le 0$; the convention understates the residual capacity after a return by $\min\{D_{\le k}, L_0\}$ and, if anything, undervalues the lever. We adopt it because it fixes the post-return state, which reduces the fresh-start value to a single number per stop; with the physical reset it becomes a function of the observed delivered volume. Section 4.5 prices the physical reset and shows that it raises the saving further, so the results we report for the depot return are conservative. Returns may be repeated: after a return the vehicle re-enters the same decision process in state $(k, 0)$, so further returns and a later handoff remain available. (The published restocking competitor is restricted to a single return, as is usual in that literature.)

The recursion (9)–(10) therefore extends to the depot-return lever with

one additional term:

$$V_k(w) \;=\; \min\big\{\, C_k(w),\;\; H_k,\;\; R_k + C_k(0) \,\big\}, \qquad k = m-1, \dots, 1, \tag{14}$$

with C_k still defined by (9); the quantity $F_k := C_k(0)$ is the *fresh-start value*, the expected cost of the remainder of the day from the reset state under optimal play with the full action set, including any further returns or a later hand-off. The optimal policy exercises, at each stop, the cheapest action whose price is below the estimated cost of continuing. Recursion (14) is the exact dynamic program for the three-action problem under the conservative reset when the increments are independent, because the law of the remainder of the day after a return then does not depend on the state that triggered it. Under factor dependence it is misspecified: the post-return remainder depends on the factor posterior, hence on the pre-return state W_k, so (k, W_k) is no longer a sufficient state once returns are allowed and the fresh-start value should be a function $F_k(W_k)$ rather than a number. The state-conditional variant BATON-CF introduced below approximates that function; as explained there, it does not recover the full information state.

Estimating the fresh-start value. The isotonic estimate $\widehat{C}_k$ is fitted on the *alive* paths at stop k, whose load states concentrate away from zero on routes that depart nearly full; reading off $\widehat{C}_k(0)$ would extrapolate the monotone fit to the edge of its support, where it is biased downward, and the fitted policy would over-return. BATON instead evaluates F_k with respect to its own fitted policy: during the backward sweep, when stop k is reached, the models and fresh-start values of all later stops are already available, so F_k is computed by simulating the reset state $(k, 0)$ forward through the training paths under the already-fitted downstream policy, nested returns valued by the already-computed F_j, $j > k$. This adds one vectorised forward pass per stop—$O(Nm)$ work—and avoids the extrapolation.

The resulting estimate $\widehat{F}_k$ differs from the optimal F_k for three reasons,

and it helps to separate three quantities: the optimal F_k; the population value F_k^{pop} of the reset state under the *fitted* downstream policy; and the estimate $\widehat{F}_k$ computed on the training paths. (i) Because the fitted downstream policy is feasible, $F_k^{\text{pop}} \geq F_k$: in population the suboptimality of the downstream policy can only raise the value. This is also the value the deployed policy will actually realise after a return, so it is the consistent price for the policy that is deployed. The inequality is a population statement; a finite-sample estimate need not respect it. (ii) The forward pass runs on the paths on which the downstream models were fitted, so $\widehat{F}_k$ differs from F_k^{pop} by an in-sample error. In-sample optimism would make it negative, but its sign is not determined a priori, because the same paths also shaped the downstream decisions. (iii) When demand is positively dependent, the suffix after a return is correlated with the state that triggered it: returns are chosen on heavy days, whose remainder tends to be heavy too, so an unconditional F_k underprices the return on exactly the days on which it is used. The appendix measures (i) and (ii) against fifty thousand independent paths per route (Table A.3): both are small fractions of the handoff price on Salhi–Nagy and Dethloff, while on the city routes the downstream difference is about one per cent of the handoff price and (iii) is material on these delivery-dominated urban routes.

The third bias suggests a remedy that stays within the method. Regressing each training path's realised post-return cost, computed from that path's own future increments, on its pre-return state gives a *state-conditional* fresh-start value $\widehat{F}_k(\cdot)$ at the cost of one more isotonic fit per stop; we report this variant as BATON-CF. It is an approximation, and two load quantities must be distinguished to see why. Let $S_k = \sum_{j \leq k} g_j$ be the cumulative observed increment, which never resets (it is the W_k of the handoff problem), and let X_k be the physical load relative to the departure load, which a return resets: after the most recent return at stop r, $X_k = S_k - S_r$ under the conservative convention. Unloading destroys no information: the operator

still observes S_k. Under independent increments the future depends on X_k alone. Under factor dependence it does not, since the posterior of the day's factor depends on S_k, so two vehicles in the same state (k, X_k) can face different predictions of the remaining demand. BATON-CF conditions the price of a return on the pre-return state, which is S_k for a first return, but its continuation models $\widehat{C}_j$ and $\widehat{F}_j$ are fitted on paths indexed by S_j and, after a return, are applied to the physical state X_j, as in BATON. It therefore reduces bias (iii) for the first return without conditioning later decisions on the full history; an exact treatment would carry the two-dimensional state (k, X_k, S_k). The empirical gains of BATON-CF should be read in that light, as a reduction of the dependence bias rather than its removal. Two further qualifications apply. Proposition 4 does not cover $F_k(\cdot)$ under dependence. A sufficient condition is that the posterior law of the remaining increments be stochastically increasing in S_k and that the cost of the remainder of the day under the downstream policy be non-decreasing in those increments, path by path; the second part need not hold for a fitted policy, so the isotonic form of $\widehat{F}_k$ is a modelling choice, which we check empirically in §4.5. And $F_k(\cdot)$ is constant under independence only in population: in finite samples the two fits differ, so BATON-CF and BATON need not take the same decisions.

Deployment selection. On routes where the return lever cannot pay—remote depots, hostile geometry—including it in the menu adds estimation noise without adding value, and the finite-sample biases above can then make the richer menu slightly worse than the handoff-only one. BATON therefore fits both the two-action and the three-action policy on the training paths, evaluates both on those same paths, and deploys whichever is cheaper. The selection uses no test data and costs one simulation pass. Because it compares in-sample costs it inherits bias (ii), which is why it is a safeguard rather than a guarantee; BATON-CF, which targets bias (iii) directly, performs as well without it (§4.5).

Algorithm 1 BATON offline phase: peak-aware backward induction over the full action set

Require: historical increments $g_k^{(s)}$, $s = 1..N$, $k = 1..m$; slack B; price schedules (H_k), (E_k), (R_k)

$W_k^{(s)} \leftarrow \sum_{j \le k} g_j^{(s)}$; $T_s \leftarrow \min\{k : W_k^{(s)} > B\}$ (∞ if none) ▷ peak, not endpoint

$\phi_s \leftarrow E_{T_s}\, \mathbf{1}\{T_s \le m\}$ ▷ cost under "never act"

for $k = m - 1$ **down to** 1 **do**

  $A \leftarrow \{s : T_s > k\}$ ▷ paths alive at stop k

  $\widehat{C}_k \leftarrow$ isotonic fit of $\{\phi_s\}_{s \in A}$ on $\{W_k^{(s)}\}_{s \in A}$

  $F_k \leftarrow$ mean cost of simulating state $(k, 0)$ forward through $\{\widehat{C}_j, F_j\}_{j>k}$ ▷ simulated fresh-start value

  for $s \in A$ **do** ▷ policy improvement at stop k

    if $H_k < \widehat{C}_k(W_k^{(s)})$ **and** $H_k \le R_k + F_k$ **then**

      $\phi_s \leftarrow H_k$ ▷ handoff: its price

    else if $R_k + F_k < \widehat{C}_k(W_k^{(s)})$ **then**

      $\phi_s \leftarrow R_k + F_k$ ▷ return: price plus estimated F_k

    end if ▷ continue: ϕ_s keeps the realised path cost

  end for

end for

return $\{\widehat{C}_k, F_k\}_{k=1}^{m-1}$

3.6. Algorithms, complexity, and reference bounds

Algorithm 1 states the offline phase for the full action set; Algorithm 2 the online loop, whose logic is also drawn in Figure 1. The offline phase performs one isotonic fit and one fresh-start evaluation per stop: $O(N \log N + Nm)$ per stop, hence $O(mN(\log N + m))$ per route in total. Measured on one core (§4.7), fitting the full policy with deployment selection takes 24 ms per route at $N = 10^3$ (median over our routes), which is what makes nightly refitting across a metropolitan fleet unremarkable. The online decision is one evaluation of a monotone piecewise-linear function—a binary search over its breakpoints—and two scalar comparisons per stop, about 1.2 μs per decision.

Three reference points calibrate the empirical results. The *clairvoyant bound* pays 0 on $\{T = \infty\}$ and $\min\{\min_{k<j} H_k,\, E_j\}$ on $\{T = j\}$—the cheap-

Algorithm 2 BATON online phase: priced-action trigger

Require: m, B, (H_k), (E_k), (R_k), fitted $\{\widehat{C}_k, F_k\}$

$W \leftarrow 0$
for $k = 1$ **to** m **do**
  serve stop k; observe d_k, p_k; $W \leftarrow W + p_k - d_k$
  if $W > B$ **then return** (EMERGENCY, E_k)
  end if ▷ physical guard: running peak
  if $k = m$ **then return** (COMPLETE, 0)
  end if
  $c \leftarrow \widehat{C}_k(W)$
  if $H_k < c$ **and** $H_k \le R_k + F_k$ **then return** (HANDOFF, H_k)
  end if
  if $R_k + F_k < c$ **then**
    pay R_k; $W \leftarrow 0$ ▷ depot return: pickups unloaded, state recorded as $W = 0$
  end if
end for

est of any pre-breach handoff or the breach itself, with a breach at the first stop leaving no decision epoch; no policy adapted to $(\mathcal{F}_k)$ that is restricted to the handoff lever pays less on any path. Because it is defined for the handoff lever, a policy with the richer menu may legitimately beat it: a depot return costs less than a handoff and does not end the route, so it can prevent a breach at a price no handoff-only policy, clairvoyant or not, can match. When BATON exceeds this bound, the excess measures the value of the second lever, not a statistical artefact. We therefore also report the *three-action clairvoyant* (oracle^3), which on each day solves by dynamic programming over (stop, last return) the cheapest sequence of returns and at most one handoff that avoids or absorbs the breach; it is a lower bound on the expected cost of every policy in this paper. We use the two words consistently: a clairvoyant is a *bound* for a class of policies when no policy in that class can do better on any day. The handoff-only clairvoyant (oracle-ho) is thus a bound for the handoff-only policies, but for the three-action policies it is only a *reference*

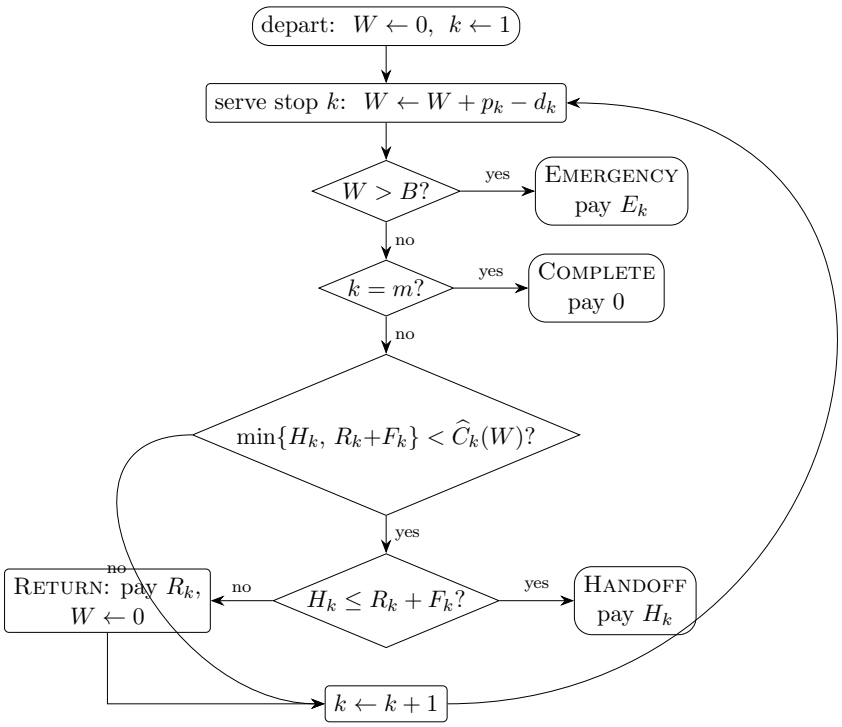


Figure 1: Online decision loop of BATON. After every stop the policy compares the estimated cost of continuing against the current prices of the two recourse levers and exercises the cheaper lever only when continuing is dearer; a depot return resets the load state and the loop proceeds on the same route.

point, which they may exceed; oracle^3 is a bound for every policy. In the results tables both appear among the reference points, which also include the high-data plug-in programs, and the captions say which of them is a bound for which policies. The *plug-in dynamic programs* replace the isotonic step by quantile-binned conditional means. Run with fifty times BATON's training data they serve as *high-data plug-in references in the* (k, W_k) *state*, one for the handoff-only problem ($\mathrm{DP}_{50\mathrm{k}}$) and one for the three-action problem ($\mathrm{DP}^3_{50\mathrm{k}}$). The latter uses the state-conditional fresh-start value $F_k(W_k)$ of §3.5, estimated on a half of the sample that is independent of the half used for the continuation values, so that it carries no in-sample optimism. These references are not exact: they are restricted to the (k, W_k) state, their bin-

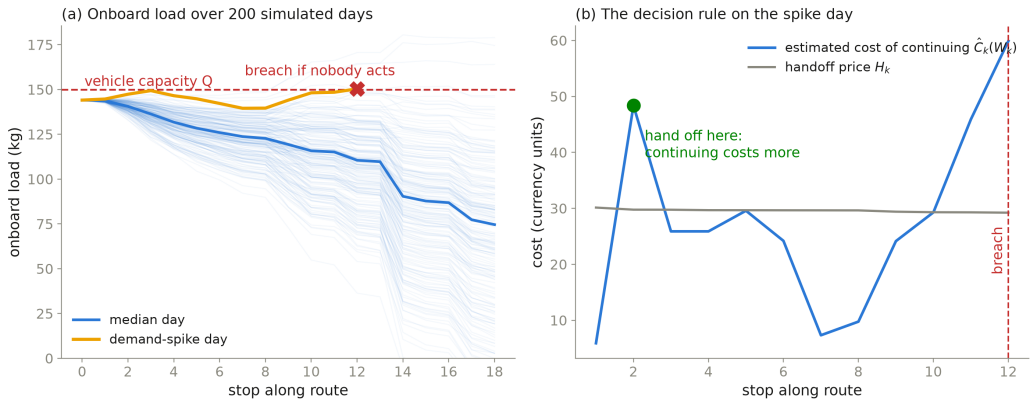


Figure 2: BATON on one 18-stop route in Hanoi. (a) Onboard load across 200 simulated days; the highlighted day breaches capacity at stop 12 if nobody acts. (b) The decision rule on that day: the estimated cost of continuing $\widehat{C}_k(W_k)$ crosses the per-stop handoff price at stop 2, where the policy hands off.

ning adds bias, and §4.7 shows BATON matching or exceeding them at large training budgets. The only exact anchor is the grid dynamic program of §4.5, which solves both problems to discretisation accuracy when demands are independent. At equal data the plug-in programs are fair equal-information competitors (DP_N, DP^3_N). We emphasise a point of method: mixed-integer programming, the exact technology of the *planning* stage, cannot compactly encode a non-anticipative multistage stopping rule, so backward dynamic programming is the correct exact anchor for the *execution* stage; we use MIP where it belongs, certifying the routing plans beneath all our experiments (§4.3).

Figures 2 and 3 make the machinery concrete on one route of the Hanoi instance introduced below: the fan of simulated load paths with a demand-spike day highlighted, the fitted continuation cost crossing the handoff price at stop 2—ten stops before the breach the reactive policy suffers—and the resulting expected cost over two thousand held-out days.

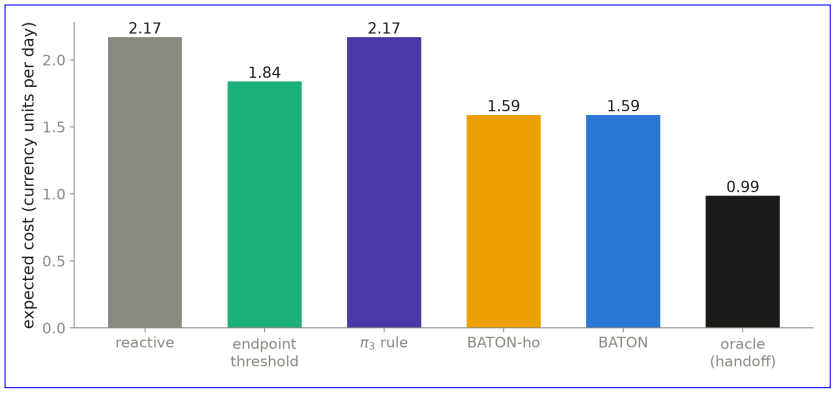


Figure 3: Expected execution cost per policy on the route of Figure 2, over 2,000 held-out days. On this route deployment selection keeps the depot return switched off, so BATON coincides with BATON-HO, although at stop 2 a return would have been slightly cheaper than the handoff; selection compares the two menus on average over the training days, not stop by stop, and here the richer menu did not pay on average. The oracle is the handoff-only clairvoyant, a bound for the handoff-only class.

4. Computational study

Our computational study is designed around a simple discipline: every claim about the policy must survive a change of environment, a change of competitor, and a change of yardstick. This section first describes the environments, competitors, and protocol (§4.1), then the headline comparison across planning regimes with its statistical analysis and tail risk (§4.2), the large-scale and real-map benchmarks with the planning-stage certification (§4.3), the comparison against learning methods and the empirical price of over-triggering (§4.4), the robustness of the conclusions to the modelling assumptions (§4.5), shared standby capacity (§4.6), data and computing budgets (§4.7), and finally the economic and spatial sensitivity analyses (§4.8). All experiments, result files, and figures are reproducible from a single repository, and every table in this paper is generated mechanically from the result files.

4.1. Instances, planners, competitors, and protocol

Instances. Three families are used. The 40 Dethloff instances (Dethloff, 2001), 50 customers each, are the classical simultaneous pickup-and-delivery benchmark. The Salhi–Nagy family (Salhi and Nagy, 1999), 14 instances of 50–199 customers, is derived from the CMT instances (Christofides et al., 1979) by the standard coordinate-ratio demand split and extends the size range. Finally, we introduce a set of instances built on the real drive networks of Ho Chi Minh City, Hanoi, New York, Paris, and Shanghai (6 km radius around the centre, 5,800–15,800 road nodes per city): customers are placed at *real shop locations* drawn from OpenStreetMap's retail layer, snapped to the nearest network node, so the spatial demand distribution is the city's actual one—market streets, malls, commercial clusters—and distances are true road distances (Boeing, 2017). Instance sizes reach 400 customers; Figure 4 shows four of the instances with their planned routes. Demands follow the benchmark protocol (gamma marginals, coefficient of variation 0.3, equicorrelated Gaussian copula $\rho = 0.6$ among the deliveries and, independently, among the pickups); the city instances use parcel-van magnitudes, with mean deliveries of 8 kg, pickup fractions uniform on $[0.2, 0.8]$, and 150 kg vehicles, which yields the long 12–25-stop routes where execution policy matters most. Because deliver-only customers are common in practice, every city instance also has two deliver-only twins in which the pickup of a random 25% or 50% of the customers is set to zero; the twins are re-planned.

Planning regimes. Routes are constructed by an adaptive large neighbourhood search planner running under six capacity-feasibility gates of increasing conservatism: a deterministic gate on the nominal load peak (Det); a sample-average CVaR gate (SAA); a Wasserstein distributionally robust gate (WDRO) (Mohajerin Esfahani and Kuhn, 2018); the quadrant-budget robust gate of Gounaris et al. (2013) (Rob-G); a budget-uncertainty gate in the sense of Bertsimas and Sim (2004) (Rob-BS); and a moment-ambiguity gate in the closed form implied by Ghosal et al. (2024) (M-DRO). With a



Figure 4: City instances on real street networks, with customers at real shop locations (Hanoi, New York, Paris, Shanghai; 100 customers each; depot marked ⋆). Note the true retail clustering—Hanoi's Old Quarter, Manhattan's commercial spine—that a uniform spatial convention misses; §4.8 quantifies its effect.

type-1 Wasserstein ball over unbounded support and a peak-load loss that is 1-Lipschitz, the worst-case CVaR equals the empirical CVaR plus the constant $\varepsilon/(1-\alpha)$ (Mohajerin Esfahani and Kuhn, 2018), so the WDRO gate is the SAA gate at a tighter threshold; we retain it as a distinct, more conservative planning regime rather than as an independent modelling paradigm. Every execution policy is evaluated on the identical plans of every gate, so planning quality never contaminates the policy comparison; the plans themselves are certified in §4.3.

Competitors. Sixteen execution policies are compared, all trained (where applicable) on scenarios strictly disjoint from the test set: the *reactive* policy that never intervenes; the incumbent endpoint-labelled threshold and its myopic variant, retained as ablations of Propositions 1 and 2; the *label-corrected tuned threshold* (peak labels, one threshold tuned on realised training cost); a *position-dependent threshold* with one load threshold per stop, tuned by coordinate descent on realised training cost, which can represent any monotone stopping boundary and therefore isolates the value of estimating that boundary by backward induction rather than searching for it (the descent starts from the best of the reactive policy, the per-stop cuts implied by the tuned global threshold and the per-stop boundaries of BATON-HO, so it is at least as good as each of them on the training days); a *two-lever* rule that combines the tuned probability threshold for the handoff with a tuned depot-return trigger—return whenever the running load exceeds a tuned fraction of the slack, repeatedly if needed—with both parameters searched jointly on realised cost, which gives a heuristic the same two levers as BATON; the three published rule-based restocking triggers π_1, π_2, π_3 of Salavati-Khoshghalb et al. (2019), adapted to the handoff lever and grid-tuned on realised cost; a *rollout* policy in the sense of Secomandi (2001), one step of policy improvement over the reactive base—which, under position-dependent prices, is exactly the myopic rule $C_k^0(W_k) > H_k$—and a *cost-scaled rollout* that hands off when $C_k^0(W_k) > \theta H_k$ with θ tuned on realised cost; a *depot-*

return restocking policy in the tradition of Yang et al. (2000) and Florio et al. (2020), with real detour prices and a tuned trigger; BATON-HO, the handoff-only restriction of our policy; BATON itself; the equal-data plug-in dynamic programs DP_N (handoff only) and DP^3_N (three actions, continuation and fresh-start values fitted on the two halves of the same 10^3 training days); and BATON-CF, the variant with a state-conditional fresh-start value. BATON-CF was introduced during the revision, after the city results had exposed the dependence bias it reduces (§3.5); we report it beside BATON rather than in its place. A reinforcement-learning policy re-implemented from Iklassov et al. (2024) is compared separately in §4.4. The two high-data programs $\mathrm{DP}_{50\mathrm{k}}$ and $\mathrm{DP}^3_{50\mathrm{k}}$ and the two clairvoyant bounds close every table as reference points. Table 1 collects the roster with sources; in every results table the reported quantity is the expected-recourse saving relative to the reactive policy, in percent, so *higher is always better*.

Protocol. For each instance, one set of 10^3 training days, one set of 2×10^3 test days and one set of 5×10^4 reference days are drawn from independent random streams whose seeds are fixed functions of the instance name, so every run of the pipeline draws identical scenarios. All routes of a plan share the same days, as they would in operation. Every data-driven policy—BATON, the thresholds, the two-lever rule, the rule-based triggers, the rollouts, the restocking trigger, DP_N, DP^3_N and the RL policy—sees exactly the same 10^3 training days; hyper-parameters of the tuned competitors are selected on realised cost over those same training days (there is no separate validation set, which, if anything, favours the tuned competitors: BATON has no continuous hyper-parameter, and its one discrete choice, the recourse menu, is made on the same days). The test days are used only for the reported evaluation. The reference programs use the 5×10^4 reference days; $\mathrm{DP}^3_{50\mathrm{k}}$ fits its continuation values on one half and its state-conditional fresh-start values on the other (sample splitting; DP^3_N does the same on the two halves of the 10^3 training days, whereas BATON uses all 10^3 days for both, so the comparison of the

Table 1: Execution policies compared in this paper. Policies marked "tuned" receive their best hyper-parameter(s) by search on realised cost over the training scenarios; BATON has no continuous hyper-parameter (its one discrete choice, the recourse menu, is made in sample). The last four rows are reference points, not competitors; the last column says for which policies each clairvoyant is a bound.

Label	Policy	Source
reactive	never intervenes; emergency on breach	baseline
endpoint $+\tau$	endpoint-label threshold (tuned), and its myopic variant	ablation (Props. 1, 2)
thr.	peak-label threshold (tuned)	threshold ablation
thr.-k	one load threshold per stop (tuned by coordinate descent)	position-dependent threshold
two-lever	tuned handoff threshold + tuned repeatable return trigger	two-lever heuristic
π_1–π_3	rule-based restocking triggers (tuned)	Salavati-Khoshghalb et al. (2019)
roll.	one-step policy improvement (myopic cost rule)	Secomandi (2001)
roll.-θ	myopic cost rule with tuned price multiplier	cost-dependent threshold
restock	depot-return restocking, tuned trigger	Yang et al. (2000); Florio et al. (2020)
RL	policy-gradient network	Iklassov et al. (2024)
DP_N, DP^3_N	plug-in dynamic programs, equal data, handoff-only / three actions	Secomandi and Margot (2009)
BATON-HO	handoff-only restriction of our policy	this paper
BATON	priced-action optimal stopping	this paper
BATON-CF	BATON with state-conditional fresh-start value	this paper
$\mathrm{DP}_{50\mathrm{k}}$, $\mathrm{DP}^3_{50\mathrm{k}}$	plug-in dynamic programs, 50× data, handoff-only / three actions	high-data reference
oracle-ho	clairvoyant per-day optimum, handoff only	bound for handoff-only policies; reference point for three-action ones
oracle^3	clairvoyant per-day optimum, three actions	bound for every policy

two reflects the allocation of the sample as well as the difference between the estimators). The two clairvoyant bounds are evaluated on the test days. Offline fitting time and online decision time are reported separately in §4.7.

Statistical analysis. The experimental unit is the instance within a planning gate: policies are compared on the plan-level expected recourse cost, the sum over the plan's routes of the mean over the 2×10^3 test days. Routes of a plan share their test days and are therefore not treated as independent observations, and the six gates re-use the same forty instances, so we analyse each gate separately ($n = 40$) rather than pooling them. For each comparison we report the mean paired difference in saving with a 95% bootstrap confidence interval, the number of instances won, the matched-pairs rank-biserial effect size, and a two-sided Wilcoxon signed-rank test with Holm's correction across all comparisons of a table. Where the comparator is "the strongest competitor", it is chosen by mean saving on the same instances; the reported interval then re-selects the comparator in every bootstrap resample, and the p-value is the largest of the p-values against each candidate competitor (an intersection-union test), so that the selection does not inflate the evidence. Route-level results for every policy, gate and instance are provided as supplementary data, and the tail of the daily plan-level bill is analysed separately.

Fleet economics. The defaults are $F_{\text{plan}} = 35$, $F_{\text{sb}} = 20$, $F_{\text{ho}} = 5$, $c_{\text{tr}} = 3$, $F_{\text{emg}} = 40$, $s_{\text{ho}} = 1.2$, $s_{\text{emg}} = 2.5$, $p_{\text{late}} = 1.5$, $p_{\text{breach}} = 10$ (currency units per vehicle-day or per event; kilometre costs $c_{\text{km}} = 0.1$), calibrated to the order of magnitude of Southeast-Asian gig-logistics rates and swept in §4.8.

Software. All planning and execution code is Python 3.11.15 (NumPy 2.4.6, SciPy 1.17.1, pandas 3.0.3); the isotonic regressions of Algorithm 1 use scikit-learn 1.9.1. The MIP certification of §4.3 solves the two-commodity flow formulation through PuLP 3.3.2 with Gurobi 13.0.2 as the primary solver and the open-source HiGHS 1.15.1 as an independent cross-check; both solvers

Table 2: Expected-recourse saving over the reactive policy (%, mean over the 40 Dethloff instances) for each planning gate and execution policy under the three-class fleet cost model; *higher is better*; the best implementable policy of each row is in bold. thr. is the label-corrected tuned threshold, thr.-k the position-dependent threshold, roll.-θ the cost-scaled rollout, two-lever a tuned handoff threshold combined with a tuned depot-return trigger that may fire repeatedly, and DP^3_N the three-action plug-in dynamic program fitted on the same 10^3 training days as BATON. π_1, π_2 and the plain rollout are dominated by π_3 and roll.-θ on every gate and are omitted; the equal-data handoff-only program DP_N is discussed in Section 4.4. BATON-CF, the state-conditional variant of Section 3.5, was introduced during the revision after the city results and is reported for completeness. The last four columns are reference points, not competitors: DP_{50k} and DP^3_{50k} are high-data plug-in programs in the (k, W_k) state for the handoff-only and the three-action problem (fifty times the training data); oracle-ho is the clairvoyant restricted to the handoff lever, a bound for the handoff-only policies but only a reference point for the three-action ones, which may therefore exceed it; oracle^3 is the clairvoyant with all three actions, a bound for every policy.

	published / tuned competitors							this paper			reference points			
Gate	π_3	roll.-θ	restock	thr.	thr.-k	two-lever	DP^3_N	BATON-HO	BATON	BATON-CF	DP_{50k}	DP^3_{50k}	oracle-ho	oracle^3
DET	17.0	24.9	5.8	24.7	26.0	26.5	27.8	26.1	27.9	**29.2**	26.4	29.5	40.7	47.2
SAA	12.6	18.4	43.7	16.3	19.8	43.9	35.4	20.1	53.7	**53.8**	22.1	56.4	43.5	78.4
WDRO	8.0	8.9	34.7	-1.3	7.2	32.4	-1.4	10.0	41.2	**41.7**	13.2	47.8	42.0	79.4
ROB-G	15.7	19.1	43.6	16.7	20.7	43.8	36.2	21.0	53.2	**53.8**	22.9	56.3	43.3	78.0
ROB-BS	11.1	18.7	42.0	16.7	20.3	42.2	34.3	20.7	52.5	**53.0**	22.6	54.9	43.1	76.6
M-DRO	9.6	10.3	37.4	0.5	8.5	35.6	-4.7	11.8	45.3	**45.6**	14.2	50.2	42.2	79.4

run single-model with a 300-second budget per instance and default tolerances. Street networks and shop locations come from OpenStreetMap via OSMnx 2.1.0 (Boeing, 2017), and the reinforcement-learning baseline is trained with PyTorch 2.14 on the same CPU. Everything else—the ALNS planner, all six gates, and every execution policy of Table 1—is implemented from the cited papers in plain Python within our repository, so all competitors share the same simulator, cost model, and data.

4.2. Headline comparison across planning regimes

Table 2 reports the central result, Figure 5 draws it, and Table 3 gives the paired statistics. Four observations organise the discussion.

First, a BATON variant has the highest saving of the implementable policies on every gate, but the margin over the strongest competitor depends on the gate. On the five conservative gates the strongest competitor is the two-

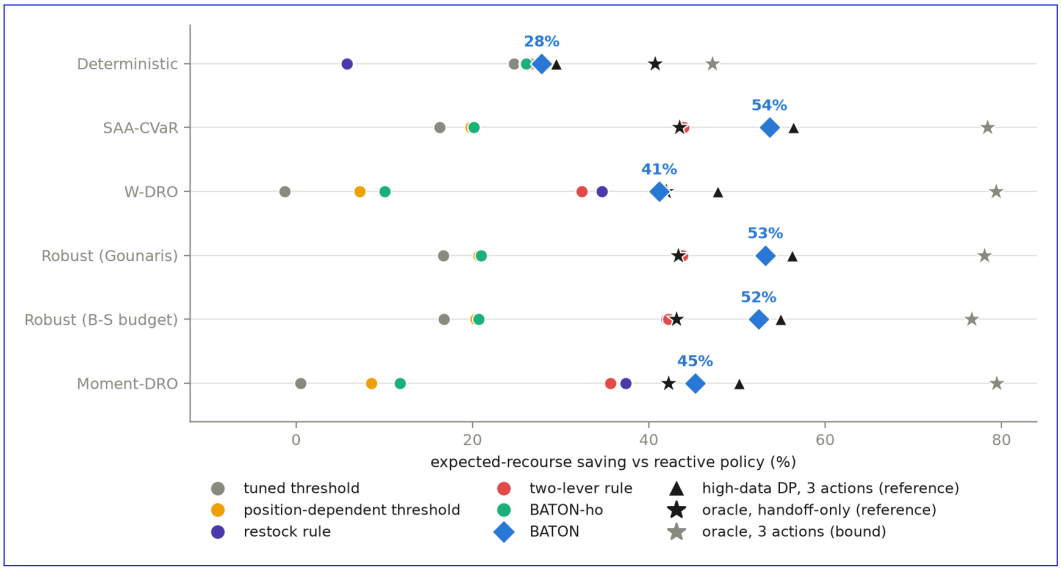


Figure 5: Expected-recourse saving by planning gate and execution policy (Table 2 drawn). BATON (diamonds) leads the plotted implementable policies on every gate (BATON-CF, not plotted, is slightly higher) and exceeds the handoff-only clairvoyant (black stars; a bound only for the handoff-only class) on four of the six gates, where plans leave enough slack for the depot-return lever to work with; the three-action clairvoyant bound (grey stars) lies above every policy.

lever rule or depot-return restocking, and BATON's mean advantage ranges from 6.5 to 10.3 percentage points; the 95% confidence interval excludes zero on each of these gates, BATON is better on at least 31 of the 40 instances, and $p_{\text{Holm}} < 10^{-3}$. On the deterministic gate the strongest competitor is the equal-data three-action dynamic program DP^3_N, and the difference is not statistically detectable ($\Delta = +0.0$ points, 95% interval [-0.4, +0.4], better on 21 of 40 instances, $p_{\text{Holm}} = 0.49$). There BATON-CF is ahead of both (29.2% against 27.9%). BATON-CF exceeds BATON on all of the six gates (by 0.1 to 1.3 points where it does); as stated in §4.1, it was introduced during the revision, so we report BATON as the headline policy and BATON-CF beside it. Against the high-data three-action reference $\text{DP}^3_{50\text{k}}$, which sees fifty times its data, BATON attains 86%–96% of its saving on the Dethloff gates (a shortfall of -1.6 to -6.6 points; §4.7 shows it closing as the history grows),

Table 3: Paired comparison of BATON with, in turn, the strongest competitor on each gate, its handoff-only restriction, and the high-data three-action reference program. The strongest competitor is not fixed in advance: it is the implementable policy other than the BATON variants with the highest mean saving among the ten competitors of Table 1 that were run on all gates. Because it is selected on the same instances, its interval re-selects the competitor in every bootstrap resample, and its p-value is the largest of the Wilcoxon p-values against the ten competitors taken one at a time (an intersection-union test of "better than every competitor", valid without a selection correction). The experimental unit is the instance ($n = 40$ per gate; the routes of a plan share their test days and are aggregated within the plan). Δ: mean difference in saving (percentage points; positive favours BATON) with a 95% bootstrap confidence interval; W: instances on which BATON is better; r: matched-pairs rank-biserial effect size; p: two-sided Wilcoxon signed-rank test, Holm-adjusted over the 18 comparisons.

		vs. strongest competitor				vs. BATON-HO				vs. DP^3_{50k} (reference)			
Gate	competitor	Δ [95% CI]	W	r	p	Δ [95% CI]	W	r	p	Δ [95% CI]	W	r	p
DET	DP^3_N	+0.0 [-0.4, +0.4]	21	+0.13	0.489	+1.8 [+1.4, +2.2]	40	+1.00	$<10^{-4}$	-1.6 [-2.2, -1.2]	0	-1.00	$<10^{-4}$
SAA	two-lever	+9.7 [+8.2, +11.1]	40	+1.00	$<10^{-4}$	+33.5 [+32.0, +35.0]	40	+1.00	$<10^{-4}$	-2.7 [-3.3, -2.2]	0	-1.00	$<10^{-4}$
WDRO	restock	+6.5 [+3.6, +9.7]	31	+0.70	$<10^{-3}$	+31.2 [+28.6, +33.7]	40	+1.00	$<10^{-4}$	-6.6 [-8.9, -4.6]	3	-0.91	$<10^{-4}$
ROB-G	two-lever	+9.4 [+7.7, +10.9]	40	+1.00	$<10^{-4}$	+32.3 [+30.5, +33.9]	40	+1.00	$<10^{-4}$	-3.1 [-3.6, -2.6]	1	-1.00	$<10^{-4}$
ROB-BS	two-lever	+10.3 [+8.1, +12.3]	40	+1.00	$<10^{-4}$	+31.8 [+30.3, +33.2]	40	+1.00	$<10^{-4}$	-2.5 [-2.9, -2.0]	2	-0.99	$<10^{-4}$
M-DRO	restock	+7.9 [+5.1, +11.1]	34	+0.89	$<10^{-4}$	+33.5 [+30.9, +36.1]	40	+1.00	$<10^{-4}$	-5.0 [-7.2, -3.1]	5	-0.84	$<10^{-4}$

and 52%–68% of the saving of the three-action clairvoyant, which knows the day's demands.

Second, the *composition* of the win changes across gates in an instructive way. On the deterministic gate, routes depart packed to capacity, most breach risk sits at the first stop where no online policy can act, and all preventive policies compress toward each other; BATON's margin over the tuned threshold there (27.9% against 24.7%) comes mostly from the trigger correction of Proposition 2, and the depot return adds 1.8 points. On every conservative gate—the two sample-based gates and all three published robust planners—plans carry slack, breaches happen mid-route, and the depot-return lever becomes live: BATON more than doubles the saving of the best handoff-only policy (for instance 53.7% against 20.1% on SAA plans) and exceeds the handoff-only clairvoyant bound (43.5%) on four of the six gates. As explained in §3.6, that bound is clairvoyant only about the handoff lever: beating it is not a paradox but a measurement of how much the second lever is worth when priced optimally. The three-action clairvoyant (78.4%

on SAA) remains well above every policy.

The savings are percentages of each gate's own reactive cost, and that cost differs by a factor of about five between gates, which explains the contrast between SAA and WDRO. Although the WDRO gate is the SAA gate at a tighter threshold (§4.1), the tighter threshold produces much safer plans: the reactive policy costs 2.9 per plan-day on WDRO plans against 16.5 on SAA plans, and at least one route breaches on 2.3% of days against 10.0%. In absolute terms BATON's expected daily cost is far lower on WDRO plans (1.9 against 8.3), but there are fewer breaches to prevent, and those that remain are harder to anticipate: a thousand training days contain few of them, which is the rare-breach regime in which BATON falls furthest short of the high-data reference (41.2% against 47.8% for DP^3_{50k}, compared with 53.7% against 56.4% on SAA). The handoff-only clairvoyant, which knows the day's demands, saves a similar share on both gates (42.0% and 43.5%), so BATON exceeds it by a wide margin on SAA plans but not on WDRO plans. (The savings are averages of per-plan percentages and the costs averages of per-plan costs, so a saving is not one minus the ratio of two average costs.)

Third, a second lever alone does not explain the gain, and neither does a richer threshold class. The two-lever rule gives a tuned threshold the depot return as well; it captures much of the three-action gain on the conservative gates (43.9% on SAA) but stays 1.4–10.3 points below BATON across the gates: with both levers available, the difference lies in *pricing* them against the cost of waiting rather than in having them. The equal-data three-action program DP^3_N prices the same actions by quantile bins; it is level with BATON on the deterministic plans, where breaches are frequent, and collapses on the conservative ones, where they are rare (35.4% on SAA, -1.4% on WDRO), because bins cannot pool information along the load axis the way the monotone projection does. For the handoff lever alone, the position-dependent threshold, with its search started from the tuned global threshold and from BATON-HO's boundaries, comes within 0.1–3.3 points of BATON-HO on every

Table 4: Tail risk of the daily recourse bill of a plan (40 Dethloff instances per gate, 2,000 test days each). CVaR_{95}: mean of the worst 5% of daily plan bills; the reactive column gives its level in currency units, the others its reduction (%, higher is better). P(emg): share of days on which at least one route of the plan suffers an emergency (%, lower is better). Best value of each group in bold.

	CVaR_{95} of daily plan bill				P(emg) per day (%)			
Gate	reactive	thr.	BATON-HO	BATON	reactive	thr.	BATON-HO	BATON
DET	481.5	23.6	**23.8**	22.3	61.7	43.5	**41.4**	**41.4**
SAA	256.0	20.1	22.2	**54.3**	10.0	6.6	6.4	**4.3**
WDRO	58.9	1.3	7.2	**37.9**	2.3	1.6	1.7	**1.3**
ROB-G	280.2	21.3	23.2	**52.2**	11.3	7.7	7.6	**5.2**
ROB-BS	302.7	21.1	22.9	**52.0**	11.6	8.3	8.1	**5.6**
M-DRO	72.2	3.0	9.3	**42.4**	2.7	1.9	2.0	**1.4**

gate (26.0% against 26.1% on DET): a stop-dependent threshold can represent the handoff boundary, and a well-started search recovers most of it. The tuned global threshold, by contrast, is negative or negligible on WDRO and M-DRO plans (-1.3% and 0.5%): with handoffs carrying their true standby cost, its interventions cost about as much as they prevent.

Fourth, the gains extend to the tail. Table 4 reports the mean of the worst 5% of daily plan-level bills and the probability that at least one route of a plan suffers an emergency on a given day. BATON cuts the CVaR_{95} of the daily bill by 22.3%–54.3% relative to the reactive policy and, on the conservative gates, lowers the daily emergency probability more than any handoff-only policy, because a depot return prevents breaches that a priced handoff would not be worth. On DET plans the emergency probability of all policies stays high, again because the risk sits at the first stop, and BATON's reduction of the CVaR_{95} (22.3%) is slightly smaller than that of the tuned threshold (23.6%) and of BATON-HO (23.8%): there the depot return lowers the mean bill but not the bill of the worst days, on which the breach happens before any return can help.

The synthetic scenarios of Table A.1 isolate the two defects of §3.3 under controlled conditions and confirm Proposition 1 at its extremal point:

on collect-then-deliver structure the endpoint-trained policy saves exactly 0.0%—it never intervenes—while the label-corrected policies save about half of the reactive cost, the option-value correction adds a further margin, and a depot return (priced flat at half a handoff) lifts BATON to 69.1%.

4.3. Scale, real maps, and certified plans

Table 5 extends the comparison to the Salhi–Nagy family and the city instances, all on DET-gate plans. On Salhi–Nagy, whose 50–199-customer routes are tight and long, the picture of the conservative Dethloff gates repeats at scale: BATON saves 53.3% of reactive cost—against 42.6% for its handoff-only restriction and beyond the handoff-only oracle (49.2%)—with the depot-return lever doing heavy work on the centrally located depots of the CMT geometry. The equal-data three-action program comes close here ($\Delta = +0.7$ points, 95% interval [+0.2, +1.2]), and BATON reaches 98% of the high-data reference.

On the city instances the return lever rarely pays. Deployment selection keeps it in the menu on only 37% of the routes, and BATON equals its handoff-only restriction (10.7% against 10.6% across 19 instances) rather than paying the estimation cost of a useless action; it leads the strongest competitor, the position-dependent threshold, by +0.5 points ([+0.1, +0.7]). We attribute the idle return lever to the demand profile and the plan rather than to the road network as such. The city routes are delivery-dominated, so their load drifts downward and rarely approaches capacity late in the day, and they are DET plans, which carry no slack: on the Dethloff DET plans the depot return also adds only 1.8 points. We could not separate the two explanations with slacker city plans, because on SAA plans for the same instances (12.1 routes per plan instead of 8.5) breaches all but vanish—the reactive policy costs 0.04 per plan-day—and no policy has anything to save. Two reference points lie above BATON on the city instances. The oracle knows the day's demands in advance; on city routes much of the breach risk comes from demand surprises at the next stop that no policy observing only the past can anticipate, which

is why its lead is large (36.8%). DP_{50k} is fitted on fifty times more data; with the return lever idle, the gap between it and Baton (12.2% against 10.7%) is a gap to a reference that has more data *and* a different estimator (binned means instead of the isotonic fit), so it cannot be attributed to the data alone. §4.7 shows that it closes, and reverses, when Baton is given a comparable history, which indicates that the amount of data accounts for most of it. The uniform-scatter twins behave the same way, with Baton and the cost-scaled rollout not statistically distinguishable ($\Delta = +0.0$, [-0.8, +0.6]).

The deliver-only twins test a feature of real demand that the benchmark protocol lacks. With a quarter or a half of the customers collecting nothing, the load rises less along the route, breaches become rarer, and every policy has less to save. Baton still saves 5.0% and 3.5%, while the tuned threshold falls to -1.3% at 50% deliver-only customers. At 25% Baton leads by +0.4 points ([+0.05, +0.62]); at 50%, where recourse is worth least, the difference from the equal-data handoff-only program is not statistically detectable ($\Delta = -0.1$, [-7.1, +2.4]) and reaches only 50% of the high-data reference, the lowest share among our settings with a non-negligible reactive cost: with breaches this rare a thousand days carry little information about them.

Figures 6 and A.1 put these aggregates back on the map, using the same rendering of the real road network on which the instances were built; every trail in both figures follows the actual shortest road path between consecutive stops, so what is drawn is what the per-stop price schedules of §3.1 price. Figure 6 replays the demand-spike day of Figure 2 on the Hanoi network. The planned vehicle (solid trail) works outward from the depot while its fitted continuation cost climbs with the accumulating pickups; at the circled stop the estimate crosses the handoff price and Baton calls the standby vehicle, which completes the dashed remainder of the route. The red cross marks where the reactive policy, facing the identical demands, instead runs the ve-

Table 5: Large-scale benchmarks under the fleet cost model (DET-gate plans unless stated; saving % vs. reactive, *higher is better*; the best implementable policy of each row is in bold; columns as in Table 2, the four columns after BATON-CF being reference points). Salhi–Nagy instances carry 50–199 customers; the city instances (100–400 customers) place customers at real OSM shop locations on the road networks of Ho Chi Minh City, Hanoi, New York, Paris and Shanghai. The uniform twins use the same cities and demands with uniformly scattered customers; the deliver-only twins set the pickup of a random 25% or 50% of the customers to zero and are re-planned. Last column: BATON minus the strongest competitor of the row (named), mean over instances in percentage points with a 95% paired bootstrap interval that re-selects the strongest competitor in every resample.

Benchmark	n	restock	thr.	thr.-k	two-lever	DP_N	DP^3_N	BATON-HO	BATON	BATON-CF	DP_{50k}	DP^3_{50k}	oracle-ho	oracle^3	Δ vs. best competitor
Salhi–Nagy	14	26.6	37.7	42.5	45.5	42.2	52.6	42.6	53.3	**53.9**	43.1	54.3	49.2	62.6	+0.7 [+0.2, +1.2] (DP^3_N)
City, real shops	19	0.3	9.8	10.2	10.1	6.8	3.4	10.6	10.7	**10.9**	12.2	12.4	36.8	37.7	+0.5 [+0.1, +0.7] (thr.-k)
City, uniform	9	0.1	11.0	10.7	11.1	7.9	4.6	11.1	11.2	**11.5**	12.2	12.5	34.3	35.4	+0.0 [-0.8, +0.6] (roll.-θ)
City, 25% deliver-only	19	-0.3	3.8	4.5	3.9	2.2	1.6	4.9	5.0	**5.1**	6.4	6.6	31.7	32.3	+0.4 [+0.05, +0.62] (thr.-k)
City, 50% deliver-only	19	-0.5	-1.3	0.4	2.0	**3.6**	2.5	3.4	3.5	3.5	7.2	7.0	40.0	40.7	-0.1 [-7.1, +2.4] (DP_N)

hicle to overflow—mid-route, far from the depot, at the position where the emergency transfer is priced highest and the largest number of downstream customers is kept waiting. Figure A.1 widens the lens from one route to an entire 200-customer plan on the same network: thirteen vehicles run simultaneously under Baton through one shared high-demand day. Seven routes hand off mid-route (circles; the standby vehicle continues on the dashed trail in the route's colour), two absorb demand spikes too abrupt for any profitable anticipative action and fall through to the emergency guard, and the remaining four complete unaided. The figure makes the operating regime of the policy concrete: recourse is not an exceptional disaster response but a routine, priced re-balancing of work across the fleet, exercised on roughly half the routes of a hard day at a total recourse bill of a few hundred currency units.

The routing plans beneath every number above were certified against mixed-integer programming bounds, using the two-commodity flow formulation of Montané and Galvão (2006) solved by Gurobi with a 300-second budget per instance. The dual bounds certify the ALNS plans within 9.34% of optimality on average (worst case 12.55%), and at equal time budgets the solver's own incumbents beat the heuristic plans on only 17/40 instances—so the execution-stage comparisons run on plans that are close to optimal and representative of what strong planning practice produces. On 12–24-customer sub-instances, where the MIP proves optimality outright, even the fast construction heuristic used for the largest instances lands within 1.2–3.6% of the optimum.

4.4. Against learning methods, the price of over-triggering, and two deliberate negative results

Because Baton is itself a learned policy, the natural challenge is a more flexible learner. We re-implemented the reinforcement-learning approach of Iklassov et al. (2024)—a policy network (two hidden layers of 64 units) over the observable state (position, load, slack, price ratios, remaining-demand

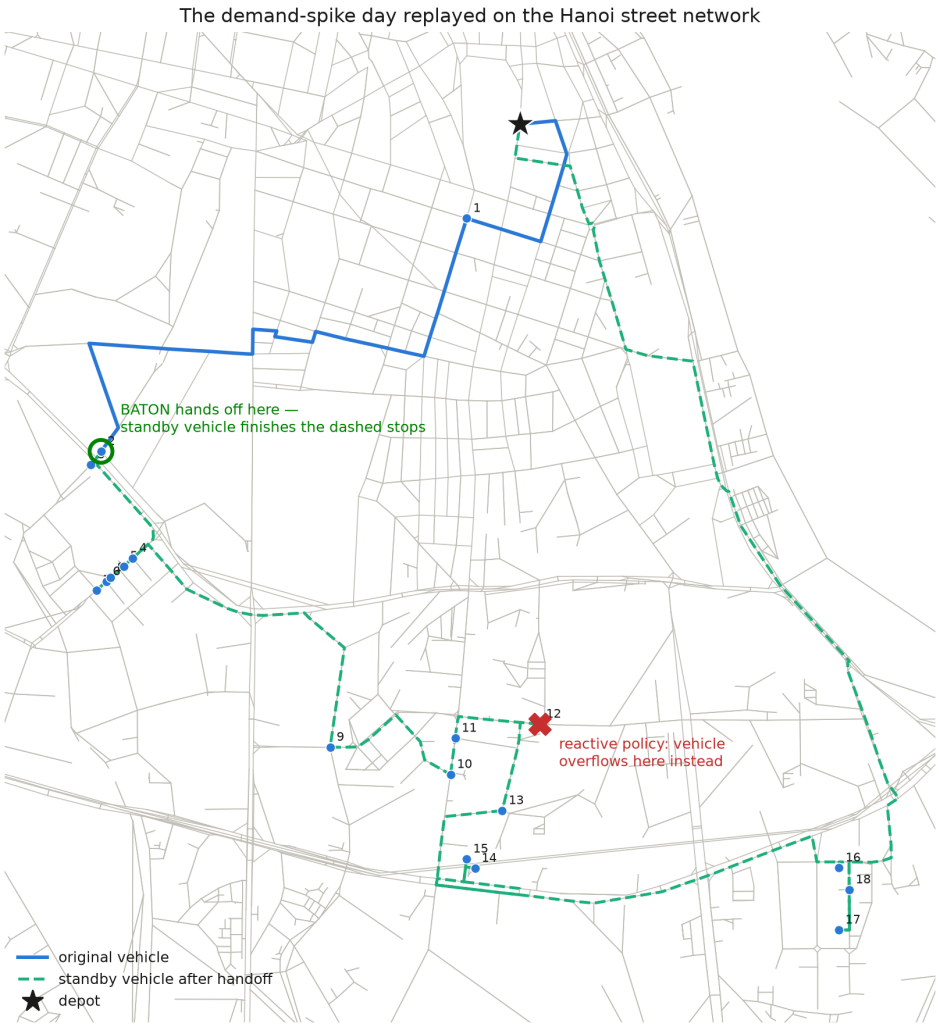


Figure 6: The demand-spike day of Figure 2 replayed on the Hanoi street network (instance HANOI-100-1; grey lines are the OSM drive network, the star is the depot, and every trail follows real shortest road paths). The planned vehicle drives the solid route; at the circled stop BATON's estimated cost of continuing exceeds the handoff price and a standby vehicle takes over the dashed remainder. Under the reactive policy the same vehicle continues and overflows at the red cross, triggering the surge-priced emergency transfer with the rest of the route still unserved.

moments), trained by REINFORCE with a moving-average baseline, Adam with learning rate 3×10^{-4} and batches of 256 days—over the identical routes and training scenarios available to BATON, evaluating both on identical held-out days. Table A.2 reports the outcome for the specification's 40 epochs and for 150 epochs under three seeds. The learned policy is respectable, clearly beating the rule-based family, but its best run saves 27.4% against BATON-HO's 32.9% on the same days (savings pooled over the 49 routes). Because the routes of an instance share their test days, we compare the two at the instance level: BATON-HO is better on 12/12 instances, by 6.0 points of reactive cost on average (95% bootstrap interval [4.4, 7.7]; one-sided Wilcoxon test over the 12 instances, $p \leq 10^{-3}$). The three seeds cluster closely and almost quadrupling the epochs barely moves the result. This does not rule out that longer training, other hyper-parameters or another architecture would do better, but for the architecture and budgets we tested it points to sample efficiency as the main reason for the gap: a policy-gradient method must discover the value structure that the backward induction computes directly, and at a thousand paths per route the regression does better, at a small fraction of the compute (§4.7).

We have compared one learning architecture, and other architectures could narrow the gap. Value-based methods such as deep Q-learning estimate the same continuation value as BATON but with a generic function class; given the monotone structure of Proposition 4 and a scalar state, we would expect them to approach, not exceed, the isotonic regression at equal data. Actor-critic methods such as PPO reduce the variance of the policy gradient and would likely close part of the gap to BATON-HO by learning a critic that plays the role of $\widehat{C}_k$. Attention- and graph-based architectures pay off when the state is a large combinatorial object—which customers remain, where the standby vehicles are—and would become relevant for partial handoffs or fleet-level coordination, where the state is no longer a scalar. In the problem studied here the learner has little to discover that the backward

induction does not compute exactly, and we therefore read the comparison as a statement about this regime rather than about reinforcement learning in general.

The price of over-triggering. Proposition 3 bounds what the myopic rule loses to the optimal one; Table 6 evaluates both sides on the Dethloff DET and SAA plans and the city plans, with both rules estimated on fifty thousand paths so that the comparison reflects the functional form rather than estimation noise. The rule labelled optimal is itself an estimate (the backward induction on those paths), so the table verifies the proposition up to that estimation error. The pathwise ordering $\sigma^0 \leq \sigma^\star$ holds on 100.0% of the test days, and the bound holds on 734/735 routes; on the 1 remaining route the regret exceeds the bound by 0.06% of that route's reactive cost, within Monte Carlo error. Pooled over all routes (a ratio of sums, not an average of route-level ratios) the myopic rule's regret is 3.8% of the reactive cost against a bound of 7.7%; the route-level average of the ratio is 5.1%. The clean-day term—the cost of handing off on days that would have ended without a breach—is of the same size as the bound (90% of it overall; on breach days the integrand of the bound can be negative, so the share can exceed 100% on a row), so unnecessary interventions are indeed what the myopic rule pays for. The breakdown answers the practical question of when a threshold is good enough. The regret is smallest on the City DET plans (1.9% of reactive cost) and on short routes of at most 12 stops (2.7%), and larger on longer routes (6.8% and 4.6% for 13–18 and 19 or more stops); on the Dethloff plans it is 3.8% on DET and 6.8% on SAA. The tuned threshold fitted on a thousand days behaves differently from the myopic rule: its excess cost over BATON-HO is 1.0% of reactive cost on DET plans, 0.4% on the city plans and 3.0% on SAA plans. Tuning moves the threshold away from the myopic break-even level, so Proposition 3 bounds the myopic rule, not the tuned threshold; the two gaps are related but not the same quantity. On tight plans and on the city routes a tuned threshold is thus a defensible simplification

Table 6: The price of over-triggering (Proposition 3), per route, on 2,000 test days. The myopic rule and the optimal rule are both estimated on 5×10^4 paths (the myopic cost-to-go by the rollout regression, the optimal one by backward induction), so that the comparison reflects the functional form rather than estimation noise; the optimal rule is therefore an estimate from those paths. Regret: excess expected cost of the myopic rule; bound: right-hand side of (13); clean-day term: $\mathbb{E}[H_{\sigma^0}\mathbf{1}\{\sigma^0 < \sigma^\star, T = \infty\}]$; all in % of the reactive cost. $P(\sigma^0 < \sigma^\star)$: share of days on which the myopic rule hands off before the optimal one; containment: share of days with $\sigma^0 \le \sigma^\star$. Last column: excess cost of the tuned threshold over Baton-ho, both fitted on 10^3 days (% of reactive cost). All columns pool the routes of a row (ratios of sums). On breach days the integrand $H_{\sigma^0} - L_{\sigma^0}$ of the bound can be negative (a handoff cheaper than the clairvoyant remainder), so the clean-day term can exceed the bound itself.

Routes	n	$\bar{m}$	regret	bound	clean-day term	$P(\sigma^0 < \sigma^\star)$	$\sigma^0 < \sigma^\star$, $T = \infty$	contain.	thr. − Baton-ho
Dethloff, Det	256	7.8	3.8	7.7	6.9	16.3	4.7	100.0	1.0
Dethloff, SAA	320	5.9	6.8	10.9	8.4	1.6	0.4	100.0	3.0
City, Det	159	17.0	1.9	4.8	5.7	1.7	0.8	100.0	0.4
all, $m \le 12$	536	5.8	2.7	5.9	5.7	5.6	1.6	100.0	0.8
all, $13 \le m \le 18$	132	15.6	6.8	12.2	9.9	12.2	3.5	100.0	1.7
all, $m \ge 19$	67	20.5	4.6	9.3	8.9	4.9	2.0	100.0	0.9

of the handoff decision; on plans that carry slack the larger loss comes from the missing depot return, which no handoff threshold can price.

Two negative results. Two further experiments were designed to fail, and did, in ways that sharpen the method's justification. First, we enriched the conditioning statistic beyond the scalar W_k with exactly the features a representation-learning critique would demand: the precision-weighted residual that is the posterior location of the day's common demand factor, the last increment, and the running maximum increment, fitted with monotonicity-constrained gradient-boosted trees. On 25 real routes the enriched policy *lost* to the scalar isotonic policy on 23, by 3.2% of cost on average, while fitting considerably more slowly (an ensemble of boosted trees per stop instead of one isotonic pass): at operational sample sizes the extra features buy more estimation variance than signal, and Proposition 4's shape constraint is doing the statistical heavy lifting. Second, the equal-data plug-in dynamic program—the same backward induction with quantile bins in place of the

isotonic projection—collapses exactly where data are scarcest relative to risk (on WDRO and M-DRO plans it saves 0.0% and 0.0%, against 10.0% and 11.8% for BATON-HO), because bins cannot pool information across the load axis the way the monotone projection can. Together the two null results locate BATON at a complexity sweet spot: one scalar statistic, one shape constraint, no continuous tuning parameter.

4.5. Robustness to the modelling assumptions

This subsection probes three assumptions behind the model: that the scalar load is a sufficient state, that all days follow one demand law, and the treatment of the post-return state.

Dependent demand and the sufficiency of the scalar load. Table 7 holds the Dethloff DET and SAA plans fixed and re-fits and re-tests every policy under five demand laws: Gaussian-copula correlations $\rho = 0, 0.3, 0.6, 0.9$ among the deliveries and among the pickups, and a route-level multiplicative day factor on top of independent marginals. Three findings stand out. First, dependence helps rather than hurts. Every implementable policy saves more as ρ grows, because correlated demand creates heavy days whose breaches can be foreseen from the running load; under independence much of the risk is idiosyncratic and cannot be anticipated. Second, the ranking holds with exceptions that we report. A BATON variant has the highest saving of the implementable policies in every row, BATON-CF in 8 of the ten. BATON itself leads the strongest competitor with a confidence interval above zero in six rows, shows no statistically detectable difference in three—the benchmark law on DET plans and the two SAA rows with almost nothing to save, discussed below—and loses one: at $\rho = 0.9$ on DET plans the equal-data three-action program is ahead (36.5% against 35.4%, $\Delta = -1.1$ points, [-1.6, -0.7]). This is the row in which the dependence bias (iii) of §3.5 is strongest, and BATON-CF, which reduces it, saves 37.6%. On DET plans BATON attains 100%, 94% and 93% of the high-data three-action saving at $\rho = 0$, 0.6

and 0.9. Third, the monotone shape survives strong dependence. On fifty thousand paths per route, the unconstrained binned estimate of the conditional expected future cost given W_k decreases significantly between adjacent bins in 0.2% of pairs under independence, where Proposition 4 guarantees monotonicity, in 0.1%, 0.1% and 0.0% of pairs at $\rho = 0.3$, 0.6 and 0.9, and in 0.6% under the day factor. The test found no significant violations beyond its false-positive level, but its power is limited: a dip of 10% of the mean continuation cost injected into one bin is detected in only 64% of the tests, and a dip of 25% in 89%. Moderate dips of the size that would matter for the policy would usually be detected, while small ones could go unnoticed. The evidence is therefore consistent with, but does not prove, a monotone continuation value outside the setting of Proposition 4, and it supports the isotonic restriction as a reasonable approximation there.

The two SAA rows are instructive. Under independence the SAA plans, calibrated for correlated demand, are so conservative that the reactive policy costs only 2.8 per plan-day and breaches are rare in a thousand training days. Out of sample, the handoff interventions of the tuned threshold and of BATON-HO then cost more than they prevent (-24.9% and -8.0%). The position-dependent threshold also loses money (-11.9%), while BATON learns almost never to act (-0.1%) and BATON-CF is marginally positive (0.2%); the best competitor, the equal-data handoff-only program, is not statistically distinguishable from BATON ($\Delta = -0.1$, [-4.6, +2.2]). Under the day factor on SAA plans BATON saves 7.4% against 5.8% for restocking, again no statistically detectable difference ([-1.6, +4.9]), and reaches only 60% of the high-data reference. These are the small-data regimes of Remark 1: with a few dozen breaches per history, backward induction propagates noise, and a policy that declines to act is the right outcome.

An exact benchmark under independence. The plug-in references $\mathrm{DP}_{50\mathrm{k}}$ and $\mathrm{DP}^3_{50\mathrm{k}}$ are estimates. When the demands are independent ($\rho = 0$), the load is a Markov chain with known transition densities, and both the handoff-only

Table 7: Robustness to the demand dependence (40 Dethloff instances, DET and SAA plans, saving % vs. reactive; plans are held fixed and policies are re-fitted and re-tested under each demand law; react.: expected daily recourse cost of the reactive policy per plan, which shows how much there is to save; best implementable policy in bold; the last two policy columns are reference points; the tuned global threshold and the handoff-only oracle are in the replication files). ρ is the Gaussian-copula equicorrelation among deliveries and among pickups; the day-factor law multiplies every demand of a day by a common lognormal factor (s.d. 0.25) on top of independent marginals. Δ: BATON minus the strongest competitor of the row (named; BATON variants excluded), mean over instances with a 95% paired bootstrap interval that re-selects the strongest competitor in every resample. Shape test (both gates pooled): viol. is the share of adjacent-bin pairs in which the unconstrained binned estimate of $\mathbb{E}[\text{future cost} \mid W_k]$, fitted on 5×10^4 paths per route with 25 bins, *decreases* significantly (one-sided, 2.5% level; the rate under independence, where monotonicity is a theorem, calibrates the test); power: share of tests that detect a dip of 10% or 25% of the mean continuation cost injected into the middle bin.

			saving %								shape test (%)		
Demand law	gate	react.	thr.-k	two-lever	BATON-HO	BATON	BATON-CF	DP^3_{50k}	oracle^3	Δ [95% CI]	viol.	pow.$_{10}$	pow.$_{25}$
$\rho = 0$	DET	176.4	10.5	13.1	10.6	**15.5**	15.4	15.5	48.6	+2.0 [+1.5, +2.1] (DP^3_N)	0.2	51	82
	SAA	2.8	-11.9	-8.3	-8.0	-0.1	**0.2**	7.4	65.3	-0.1 [-4.6, +2.2] (DP_N)			
$\rho = 0.3$	DET	180.2	19.1	20.2	19.3	21.9	**22.5**	22.7	47.6	+1.2 [+0.9, +1.6] (DP^3_N)	0.1	58	89
	SAA	8.0	6.5	29.0	7.9	36.3	**36.6**	40.9	77.5	+6.5 [+4.7, +8.4] (restock)			
$\rho = 0.6$ (bench.)	DET	179.2	26.0	26.5	26.1	27.9	**29.2**	29.5	47.2	+0.0 [-0.4, +0.4] (DP^3_N)	0.1	72	97
	SAA	16.5	19.8	43.9	20.1	53.7	**53.8**	56.4	78.4	+9.7 [+8.2, +11.1] (two-lever)			
$\rho = 0.9$	DET	177.5	34.0	33.8	34.1	35.4	**37.6**	38.0	47.0	-1.1 [-1.6, -0.7] (DP^3_N)	0.0	88	97
	SAA	25.9	32.9	54.4	33.0	64.5	**65.9**	67.7	78.5	+10.2 [+8.8, +11.6] (two-lever)			
day factor, $\rho = 0$	DET	172.2	9.7	11.8	9.8	**14.2**	**14.2**	14.2	48.1	+2.3 [+1.4, +3.1] (two-lever)	0.6	47	78
	SAA	5.2	-12.7	-2.2	-6.9	7.4	**8.0**	12.4	70.2	+1.6 [-1.6, +4.9] (restock)			

program (9)–(10) and the three-action program (14) can be solved exactly, up to the discretisation of the load, by numerical convolution of the Gamma demand densities on a grid of 1,500 load cells per route—the fixed-sequence dynamic programming of Minis and Tatarakis (2011), applied to our two levers. Table 8 compares the policies with these exact optima on all routes of the Dethloff DET and SAA plans; the simulated and the exact reactive costs agree to within 0.39%. That agreement checks only the reactive value, so we also re-solved both programs on grids of 3,000 and 6,000 cells and with a wider truncation of the load range and of the demand supports. Values of single routes move by up to 2.91% between 1,500 and 6,000 cells (median 0.4%), but the discretisation affects the reactive, handoff-only and three-action values alike, and the pooled exact savings of Table 8 change by at most 0.01 percentage points across all grids and truncations. On DET plans BATON-HO attains 93% of the exact handoff-only saving and BATON 94% of the exact three-action saving (12.6% against 13.4%), the same share as $\mathrm{DP}^3_{50\mathrm{k}}$. On the slack SAA plans, where breaches are rare under independence and there is almost nothing for the handoff to do (the exact handoff-only saving is 0.4%), BATON attains 59% of the exact three-action saving against 86% for $\mathrm{DP}^3_{50\mathrm{k}}$: this is the rare-breach regime in which a thousand days are too few, and where the high-data reference, with fifty times the data, does markedly better. (Savings in Table 8 pool the route costs; the instance-averaged savings of Table 7 weight the many SAA plans with almost no reactive cost equally and are therefore lower.)

Known day types. The model treats days as exchangeable, whereas real demand has seasons, weekdays and promotions. Table 9 makes one day in five a promotion day, with mean deliveries 15% and mean pickups 35% above normal—returns surge after promotions—and compares three ways of fitting the policy on the same history of 10^3 days. A pooled fit ignores the day type; day-type-specific fits split that history by type (about 803 normal and 197 promotion days), as an operator can when promotions are scheduled; a

Table 8: Exact optimum under independent demands ($\rho = 0$; 40 Dethloff instances, all routes of the DET and SAA plans). The exact values solve the dynamic programs over (k, W_k) by numerical convolution of the Gamma demand densities on a grid of 1,500 load cells, for the handoff-only menu and for the three-action menu with the conservative reset. Policies are fitted on 10^3 days and evaluated on 2×10^4 test days; MC err.: relative difference between the simulated and the exact reactive cost. Re-solving on grids of 3,000 and 6,000 cells, or with a wider truncation, changes the pooled exact savings by at most 0.01 points. Savings are % of the reactive cost, pooled over routes; the last three columns give BATON-HO's share of the exact handoff-only saving and BATON's and DP^3_{50k}'s share of the exact three-action saving (–: the exact saving is below one point, so the share is not meaningful).

			MC	exact optimum											share of exact (%)		
Gate	inst.	routes	err. (%)	ho	3-act.	thr.	thr.-k	two-lever	DP^3_N	BATON-HO	BATON	BATON-CF	DP_{50k}	DP^3_{50k}	BATON-HO	BATON	DP^3_{50k}
DET	40	263	0.39	9.6	13.4	8.5	8.9	11.2	10.7	9.0	12.6	12.6	9.4	12.6	93	94	94
SAA	40	385	0.22	0.4	10.0	-3.8	-3.5	5.0	-0.1	-2.3	5.9	6.3	0.1	8.6	–	59	86

stale fit uses the normal days only and is applied unchanged on promotion days. On promotion days BATON saves 29.7% with the day-type-specific fit, 29.4% with the pooled fit and 28.3% with the stale fit. Over all days the day-type-specific fits are no better than the pooled one: pooled over plans the savings are equal (27.6% against 27.6%), and averaged over plans the split fits are slightly worse (-0.5 points, with a 95% bootstrap interval of [-0.7, -0.3] that resamples instances and keeps the two plans of an instance together; +0.1 points on DET plans, [-0.1, +0.3], and -1.1 on SAA plans, [-1.5, -0.8]), because what they gain on promotion days they lose on normal days by fitting on fewer paths, and the stale fit loses -1.3 points per plan ([-1.7, -1.0]) against pooling. The pooled policy copes with the mixture because its trigger conditions on the observed load itself: a promotion day reveals itself through a heavier running load, and the monotone continuation value prices that load whatever the day type. At this sample size the practical rule is therefore to pool the history rather than to split it, and to fit per day type only when each type has a history of its own that is long enough. The tuned threshold behaves the same way at a lower level. The scope of this experiment is narrow. Each day is drawn independently from a fixed mixture of the two types, so the days remain exchangeable: the experiment

Table 9: Known day types (40 Dethloff instances, DET and SAA plans; saving % vs. reactive on the same days, pooled over plans as a ratio of sums, with the "all" column weighting normal and promotion days 0.8/0.2). One day in five is a promotion day with mean deliveries ×1.15 and mean pickups ×1.35. Pooled: one fit on a history that mixes both day types; day-type-specific: separate fits, the day type being known in advance (promotions are scheduled), each fitted on the days of its type in the same pooled history (about 800 normal and 200 promotion days), so that the comparison isolates the day-type information from the sample size; stale: fitted on the normal days of that history only and applied unchanged on promotion days. The difference between day-type-specific and pooled BATON over all days is reported in the text, for each gate and pooled over both, with 95% bootstrap intervals that resample instances (keeping the two plans of an instance together). Days are drawn independently from a fixed mixture of the two types, so they remain exchangeable: the experiment concerns known day types, not temporal drift or serial dependence.

	tuned threshold			BATON		
Fitting strategy	normal	promotion	all	normal	promotion	all
pooled fit (day type ignored)	21.2	22.7	21.7	26.8	29.4	27.6
day-type-specific fits	21.2	22.4	21.6	26.6	29.7	27.6
stale fit (normal days only)	21.2	22.2	21.5	26.6	28.3	27.1

shows how the policy uses, or does without, day-type information and what splitting a limited history costs. It does not establish robustness to temporal drift or to serial dependence between days, which we have not studied.

The fresh-start value and the post-return state. Table A.3 (in the appendix) examines the depot-return machinery of §3.5 on the three settings where it matters most or least: Salhi–Nagy, where returns are used on about a fifth of the days; Dethloff SAA plans; and the city routes, where they rarely pay. To expose the three-action fit, all columns are computed *without* deployment selection. The first two columns separate the estimands of §3.5. The in-sample effect $\widehat{F}_k - F_k^{\text{pop}}$ is +0.40% of the handoff price on Salhi–Nagy (95% interval [+0.2, +0.6]) and -0.01% on Dethloff ([-0.05, +0.04]): small, and on these routes not the downward optimism one might expect. The difference $F_k^{\text{pop}} - F_k^{\text{ref}}$ between the fitted policy's population value and the approximate reference is +0.09% ([-0.5, +0.6]) and +0.01% ([+0.00, +0.02]). Since F_k^{ref} is itself an estimate from a binned program, this difference combines the

suboptimality of the downstream policy with the error of the reference and does not identify either. On the city routes it is negative, -1.09% ([-1.7, -0.5]): the fitted downstream policy does better than the reference there, a direct sign that the reference is only approximate. We therefore draw only the conclusion the data support: replacing $\widehat{F}_k$ by F_k^{pop} or by F_k^{ref} changes the saving by less than one point (on Salhi–Nagy 52.6% as fitted, 52.6% and 52.2%; the change from the reference is -0.3 points, [-0.77, -0.02]). Bias (iii) is different. On the delivery-dominated city routes the three-action fit without selection saves only 5.5%, below the handoff-only policy (9.0%): the running load drifts downward there, a return is chosen only on unusually heavy days, and an unconditional fresh-start value under-prices it on exactly those days. The state-conditional fresh-start value reduces the effect (9.2%) and slightly improves the other two settings as well, so BATON-CF performs about as well as BATON *with* deployment selection without needing it. The column marked "physical reset" prices the exact post-return state $x_k = -\min(D_{\le k}, L_0)$, the undelivered cargo (§3.5), instead of the conservative reset to zero. It raises the saving to 58.2% on Salhi–Nagy and 54.6% on Dethloff: the conservative convention of the main tables understates what the depot return is worth, so the three-action results we report are conservative. For deployment we recommend the physical reset; the main tables keep the conservative convention for comparability with the submitted version and with the restocking competitor, which uses the same reset. (The physical reset can exceed $\mathrm{DP}^3_{50\mathrm{k}}$, which is computed under the conservative convention and is therefore a reference for that convention only.)

Monotonicity of the state-conditional fresh-start value. The binned test applied to C_k above, applied to the realised post-return cost under BATON-CF's fitted downstream policy as a function of the pre-return state (5×10^4 paths per route, Dethloff DET and SAA plans), finds significant decreases in 2.2% of adjacent-bin pairs under independence, where $F_k(\cdot)$ is constant in population and the rate therefore measures the false-positive level of the test, and in 0.2%

and 0.2% of pairs at $\rho = 0.6$ and 0.9. The data give no sign that the isotonic form of $\widehat{F}_k$ is wrong, with the limited power discussed above for C_k.

4.6. Shared, capacitated standby capacity

The default model treats standby capacity as pay-per-use from a pool large relative to one plan (§3.1); each route then decides alone. When instead a fixed number S of standby vehicles is reserved for the day, their holding cost is paid whether or not they are used, a handoff costs only its marginal price $H_k - F_{\mathrm{sb}}$, and the routes sharing the pool compete for it. We simulate this coupling exactly for two pool scopes: one pool per plan, and a metropolitan pool shared by the DET plans of all ten instances of a Dethloff class (about 66 routes). On every test day the routes run simultaneously, handoff requests are served first come, first served in the order of the time at which they arise along the routes, and a route whose request is refused decides at the refusal stop between continuing and a depot return, and continues under the handoff-free menu fitted by the same backward induction. Two policies are compared. The *naive* policy decides every route at the marginal price and ignores the cap. The *shadow-priced* policy decides every route at the price $H_k - F_{\mathrm{sb}} + \lambda$, where the shadow price $\lambda \geq 0$ of a standby vehicle is chosen, for each pool and pool size, on the training days; it decouples the routes, which can then still be decided one at a time. We also report the two fixed prices $\lambda = F_{\mathrm{sb}}$ (routes decide as under pay-per-use) and $\lambda = \infty$ (routes never hand off). Rather than fix the holding cost of a reserved vehicle, we report the recourse cost for every pool size and derive the break-even holding cost h_1 of the first reserved vehicle—the reduction in expected daily recourse cost that it buys—together with the cost-minimising pool size at holding costs of 20 (the standby day rate) and 5 per vehicle-day.

Table 10 gives three answers. The first concerns the holding cost. A reserved vehicle takes over at most one route per day, so at a holding cost equal to the standby day rate a used reservation costs what a pay-per-use handoff costs and an idle one is pure loss: at $h = 20$ no reserved pool beats

pay-per-use in any row (for example 162.5 against 135.0 per Dethloff DET plan). Below the day rate reservation pays where handoffs are frequent. The first reserved vehicle for a Dethloff DET plan is worth 19.9 per day, and at $h = 5$ a pool of about 4.2 vehicles lowers the plan's daily cost from 135.0 under pay-per-use to 119.0, on 40/40 plans; on the metropolitan pools the reserved optimum (27.2 vehicles) costs 1109.2 against 1349.9 under pay-per-use, on 4/4 pools. On the slack SAA plans and on the city plans, which need few handoffs (first vehicle worth 0.1 and 2.6), reservation does not pay even at $h = 5$. Pay-per-use from a shared pool, the default model, is therefore the right assumption when standby capacity is bought at its day rate, and a reserved pool is worth considering when capacity can be held for well under that rate. (Under pay-per-use, the pool that covers 95% of days, a by-product of the policy, has a median of 3.5 and a maximum of 7 vehicles per plan for BATON on DET plans and a median of 0 over all six gates, where the depot return does much of the work.) The second answer concerns the coupling. Given a cap, deciding each route with a shadow price beats ignoring the cap wherever the cap binds: on the Dethloff DET plans by 4.2% with one reserved vehicle, on the city plans by 5.1%, and on the metropolitan pools by up to 9.4% (at $S = 19$), because a naive route requests handoffs it may be refused and, once refused, has already let its load climb past the point at which a return would have paid. On a metropolitan pool the first vehicle is worth little (6.6 per day): served first come, first served among some 66 routes, it goes to whichever route asks first rather than to the one that needs it most, and the shadow price chosen on the training days uses it only sparingly. The third answer concerns the price. The price chosen on the training days is high for small pools (about 40 on the metropolitan pools at $S = 2$, above the day rate) and falls as the pool grows (3.3 at $S = 40$). Neither fixed price does as well: pay-per-use pricing ($\lambda = F_{\mathrm{sb}}$) comes within a few per cent of the shadow price but stays above it at every pool size, and $\lambda = \infty$ gives up the handoff lever. The shadow price decouples the routes, so the policy is

Table 10: Shared, capacitated standby pool: expected daily recourse cost per pool (currency units, excluding the holding cost of the reserved vehicles). A pool serves the routes of one plan, or, in the metro rows, the DET plans of all ten instances of a Dethloff class (CON3, CON8, SCA3, SCA8). Pay-per-use: the default model (a handoff is billed at its full price H_k and capacity is unlimited). Reserved pool of S vehicles: a handoff costs its marginal price $H_k - F_{\text{sb}}$, requests are served first come, first served in the order of their time along the routes, and a route whose request is refused chooses at the refusal stop between continuing and a depot return. Naive: every route decides at the marginal price and ignores the cap; shadow: every route decides at $H_k - F_{\text{sb}} + \lambda$, with λ chosen on the training days from $\{0, 2, 5, 10, 20, 40, \infty\}$; the two fixed-price columns use $\lambda = F_{\text{sb}}$ (pay-per-use pricing) and $\lambda = \infty$ (never hand off) at $S = 1$. Gain: naive minus shadow at $S = 1$ with a 95% bootstrap interval over pools. h_1: break-even holding cost of the first reserved vehicle (cost at $S = 0$ minus cost at $S = 1$, shadow pricing); a reservation pays when the vehicle-day costs less than h_1. $S^\star$: mean cost-minimising pool size at a holding cost of 20 (the standby day rate) and of 5 per vehicle-day.

			pay-per-	$S = 0$		$S = 1$				gain		$S^\star$ at $h =$	
Benchmark	pools	routes	use	naive	shadow	naive	shadow	$\lambda{=}F_{\text{sb}}$	$\lambda{=}\infty$	at $S{=}1$ [95% CI]	h_1	20	5
Dethloff, DET	40	6.5	135.0	176.5	167.7	154.3	147.8	149.7	167.7	+6.4 [+4.6, +8.5]	19.9	1.1	4.2
Dethloff, SAA	40	9.5	8.3	8.3	8.3	8.2	8.1	8.3	8.3	+0.1 [+0.02, +0.08]	0.1	0.0	0.0
City, DET	19	8.4	33.8	41.5	37.1	36.4	34.5	35.6	37.1	+1.8 [+0.2, +4.0]	2.6	0.0	0.1
metro pools, DET	4	65.5	1349.9	1765.4	1677.3	1721.7	1670.7	1710.8	1677.3	+51.0 [+17.6, +84.5]	6.6	15.5	27.2

still fitted one route at a time, and the depot return is the lever that makes a small or empty pool tolerable.

4.7. Data and computing budgets

Every data-driven policy in the preceding tables was fitted on the same 10^3 training days per instance; only the two reference programs saw more. Table A.4 and Figure 7 vary the history from 10^2 to 2×10^4 days per route, on the same test days, for BATON, the label-corrected tuned threshold and the equal-data dynamic program.

BATON beats the tuned threshold at every history length we tested. With only 100 days per route it saves 25.8% on DET plans and 41.7% on SAA plans, where the threshold saves -2.4% and the equal-data dynamic program -1.6%. The isotonic projection pools information along the load axis, whereas quantile bins need many paths per bin before their estimates stabilise, which is why DP_N is roughly level with the threshold on short histories (behind it on two of the three benchmarks, ahead on the third). As the history

grows, BATON approaches the high-data references. On the SAA plans it reaches 56.3% at 2×10^4 days against 56.4% for $\mathrm{DP}^3_{50\mathrm{k}}$, and on the city routes it slightly exceeds the binned references (12.9% against 12.2%), which are estimates themselves rather than exact optima. The gaps in Table 2 between BATON and the references therefore shrink with the amount of data and do not point to a structural shortfall of the policy; since the references also use a different estimator, this experiment cannot attribute the gaps to data alone. The clairvoyant oracle's lead, by contrast, does not shrink with data: it is the value of knowing the future.

Table A.5 separates offline from online computation on one CPU core. Offline, fitting BATON-HO takes 7 ms per route and the full policy, including the fresh-start valuations and deployment selection, 24 ms. Tuning re-simulates the training days for every candidate parameter, so the grid-tuned competitors cost from less (the π_3 rule and the restocking trigger, with short grids) to considerably more (the position-dependent threshold and the two-lever rule). Online, every cost-comparison policy makes one lookup in a monotone piecewise-linear function and two comparisons per stop, about 1.2 μs per decision through the fitted breakpoints. The reinforcement-learning baseline of §4.4 is the exception: on the same CPU its training takes 1–3 minutes for the 49 routes of Table A.2, against 1.8 seconds for fitting BATON on the same routes, so the comparison of Table A.2 is, if anything, generous to the learner in computing budget and identical in data.

4.8. Economic and spatial sensitivity

A recourse policy earns its keep only if its advantage survives changes in the prices that define it. Table 11 and Figure 8 sweep the four managerial cost levers one at a time around the defaults: the standby day rate, the emergency callout, the surge multiplier, and the per-customer lateness price, and then add a fixed fee to the depot return. The savings in the table are relative to the reactive cost of the same configuration, which itself moves with the emergency prices, so a larger saving does not mean a cheaper operation.

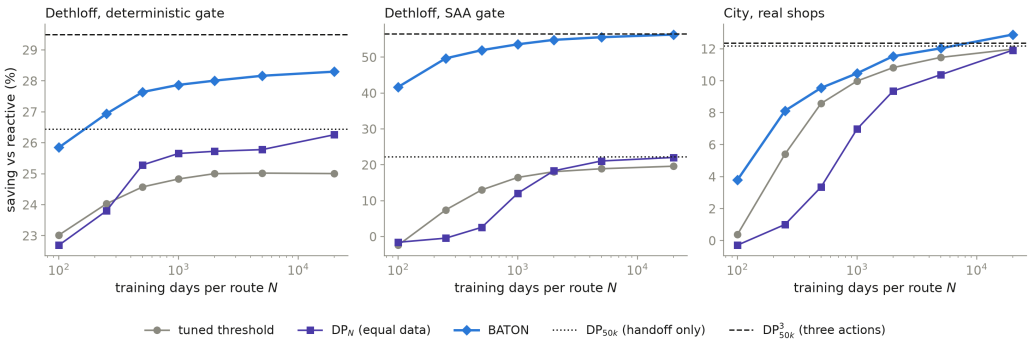


Figure 7: Saving over the reactive policy as a function of the number of training days per route (log scale), on identical test days. Horizontal lines: the high-data handoff-only and three-action plug-in programs fitted on 5×10^4 days.

Cheaper standby and cheaper emergencies both lower BATON's expected daily cost per plan, from 52.7 at the defaults to 45.8 with $F_{\text{sb}} = 10$ and to 48.3 with $F_{\text{emg}} = 25$ (and dearer prices raise it, to 60.9 and 57.7). They move the *saving* in opposite directions for a different reason: cheap standby makes the interventions cheaper while the reactive cost stays at 78.4, whereas cheap emergencies lower the reactive cost itself (to 64.2), leaving less to save; dear emergencies raise it (to 97.3) and with it the share that intervention can recover. The last two columns of Table 11 give these absolute costs. BATON leads in every configuration except the depot-return fee of 30, where it equals its handoff-only restriction; the instructive content is *how*. When standby capacity is dear ($F_{\text{sb}} = 35$), the tuned threshold collapses to 8.8%—its handoffs now barely pay for themselves—while BATON holds 42.6% by shifting its mix toward depot returns, ending far above the handoff-only clairvoyant (24.8%), which is a bound only for the handoff-only class. When lateness is expensive, depot returns lose their appeal and the mix shifts back toward handoffs. Another row sets the standby rate so high ($F_{\text{sb}} = 60$) that a handoff costs more than an emergency at late stops, violating Inequality (7). Nothing in the theory depends on that inequality (Assumption 2), and the policy behaves accordingly: it stops handing off where the handoff is dominated, relies on

Table 11: Sensitivity of the recourse saving (%, *higher is better*; best non-clairvoyant value per row in bold) to the fleet-economics parameters, one factor at a time around the defaults (12 Dethloff instances, DET and SAA gates; the baseline row is computed on the same 12 instances). two-lever: tuned handoff threshold plus tuned repeatable depot-return trigger. In the $F_{\text{sb}} = 60$ row the standby rate exceeds the emergency price at late stops, so Inequality (7) fails there. The last rows add a fixed fee (dwell, loading-dock time) to every depot return on top of its detour, transfer and lateness costs. The savings are relative to the reactive cost of the same configuration, which itself changes with the emergency prices; the last two columns therefore give the expected daily recourse cost per plan of the reactive policy and of BATON in currency units (*lower is better*). Savings are averages of per-plan percentages and costs averages of per-plan costs, so a saving is not one minus the ratio of the two cost columns.

	saving vs. reactive (%)						daily cost	
Configuration	restock	threshold	two-lever	BATON-HO	BATON	oracle-ho	reactive	BATON
baseline	28.2	24.9	41.8	29.0	**48.0**	44.5	78.4	52.7
cheap emergencies (F_{emg}=25)	23.4	13.0	33.5	18.0	**40.1**	31.0	64.2	48.3
dear emergencies (F_{emg}=60)	32.9	36.0	50.1	39.6	**55.2**	55.3	97.3	57.7
cheap standby (F_{sb}=10)	28.2	37.5	46.4	41.7	**52.7**	57.7	78.4	45.8
dear standby (F_{sb}=35)	28.2	8.8	36.6	13.1	**42.6**	24.8	78.4	60.9
low SLA price (p_{late}=0.5)	33.1	22.3	44.3	26.4	**49.0**	42.0	72.2	50.1
high SLA price (p_{late}=3)	23.6	28.5	40.0	32.3	**47.0**	47.6	87.7	55.8
mild surge (s_{emg}=1.5)	27.4	20.6	38.5	24.8	**44.7**	40.3	69.7	50.2
heavy surge (s_{emg}=4)	29.9	30.4	46.2	34.1	**52.0**	49.5	91.4	56.2
standby above emergency (F_{sb}=60)	28.2	1.4	31.1	2.1	**36.8**	5.0	78.4	68.8
depot-return fee 5	21.6	24.9	34.8	29.0	**42.4**	44.5	78.4	53.8
depot-return fee 10	16.5	24.9	30.3	29.0	**37.6**	44.5	78.4	54.5
depot-return fee 20	8.5	24.9	25.5	29.0	**30.2**	44.5	78.4	55.6
depot-return fee 30	3.9	24.9	24.7	**29.0**	**29.0**	44.5	78.4	55.7

depot returns, and saves 36.8%, while the tuned threshold saves 1.4% and the handoff-only restriction 2.1%. The last four rows charge a fixed fee for every depot return on top of its detour, transfer and lateness costs—loading-dock time, a depot slot, the driver's dwell. The value of the return lever, BATON's saving over BATON-HO, falls from 19.1 points without a fee to 0.0 points at a fee of 30, and the lever stops paying (adds less than 0.1 points) at a fee of 29 currency units; for comparison, at the start of a route on these instances a handoff costs roughly 30 to 55 and a return, before the fee, roughly 15 to 20. Where returns carry such fixed costs the two-lever advantage shrinks accordingly, and deployment selection falls back to the handoff-only menu. No parameter changes across these worlds; the policy simply re-prices its levers.

The spatial-structure experiment closes the study. For each city instance

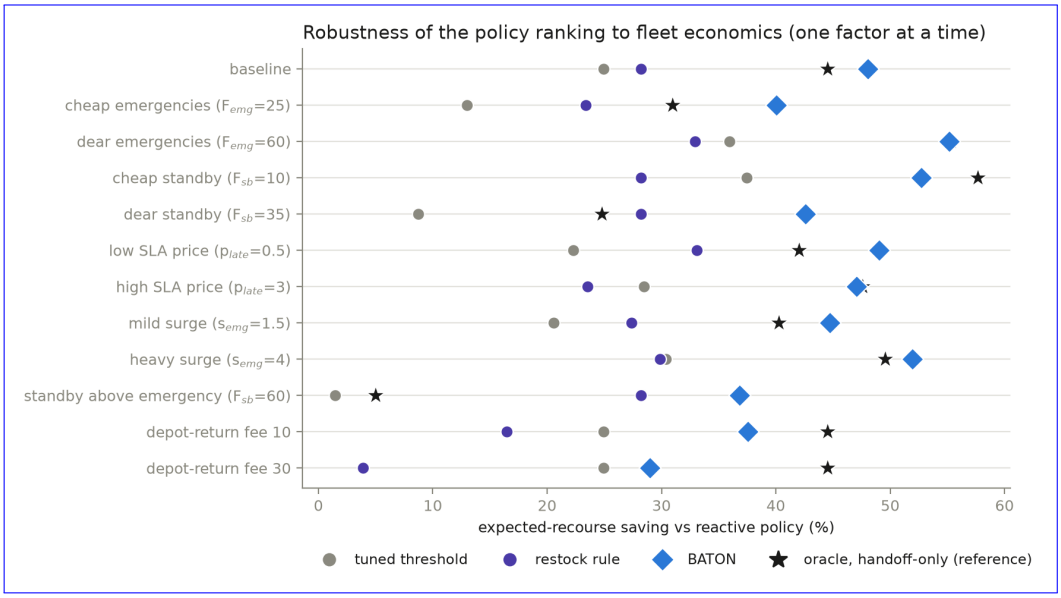


Figure 8: Sensitivity of the policy ranking to the fleet economics, one factor at a time. BATON (diamonds) leads or, with a depot-return fee of 30, equals its handoff-only restriction in every configuration, re-balancing between handoffs and depot returns as the relative prices move.

we generated a twin with identical demands and identical seeds but customers scattered uniformly over the street network rather than placed at real shops; Figure 9 shows the pair for the 200-customer Hanoi instance, and the visual difference is the experiment: real retail concentrates around the commercial core while the uniform convention spreads demand into residential and peripheral blocks. The real retail clustering shortens optimal routes by 25.9% on average—uniform spatial conventions materially overstate travel in urban simultaneous pickup-and-delivery —while BATON leads the strongest competitor, or is not statistically distinguishable from it, on both twins, with small margins in both (Table 5). The policy conclusions of this paper do not depend on where customers stand; the absolute economics do, which is precisely why we report both.

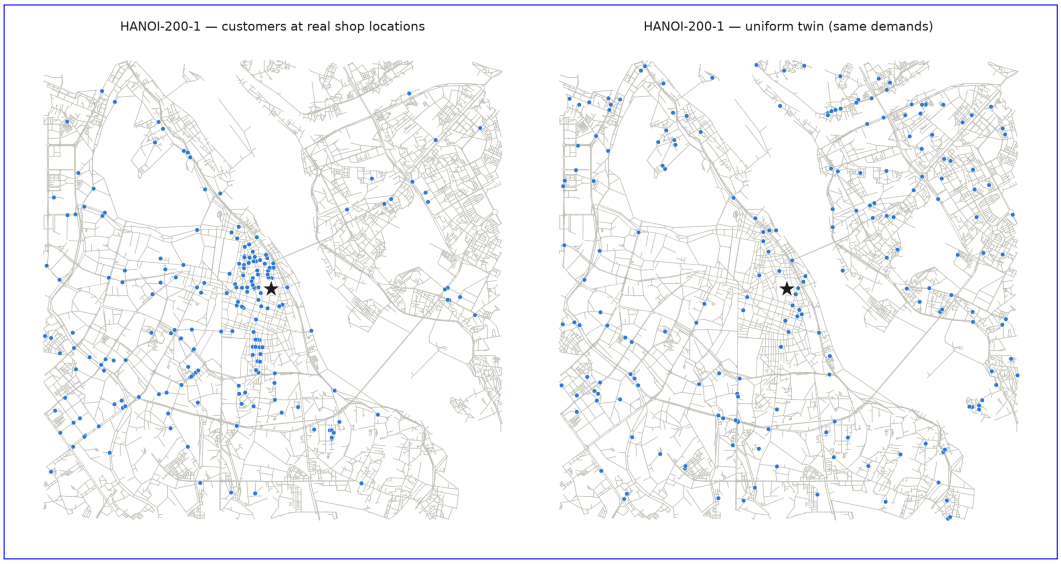


Figure 9: The spatial-structure twins for HANOI-200-1 on the OSM drive network (star: depot). Left: the 200 customers at real OpenStreetMap shop locations, clustered along the retail corridors around the commercial centre. Right: the uniform twin—identical demand draws, customers scattered uniformly over the same street network. Real clustering shortens optimal routes by 25.9% on average; BATON leads the strongest competitor, or is not statistically distinguishable from it, on both twins.

5. Managerial insights and conclusions

This paper set out from an operational nuisance—mid-route capacity breaches in simultaneous pickup-and-delivery—and arrived at a prescription for execution-stage recourse: *price the levers, estimate the cost of waiting, and compare*. The resulting policy, BATON, is a data-driven optimal-stopping rule that fits in milliseconds per route from ordinary historical demand logs, carries no continuous tuning parameter, and accommodates the recourse menu an operation actually offers. On the Dethloff benchmark under six planning regimes, against competitors spanning the rule-based, threshold, two-lever, restocking, rollout and dynamic-programming traditions (and, on a separate route set, reinforcement learning), BATON or its state-conditional variant delivered the lowest expected execution cost of every implementable policy, with margins of 6.5–10.3 points on the conservative regimes and no

statistically detectable difference from an equal-data dynamic program on deterministic plans. It reached 86%–96% of the saving of a high-data plug-in program on those regimes and, under independent demand, 94% of the exact optimum on deterministic plans and 59% on slack ones; across all benchmarks and demand laws with a non-negligible reactive cost the share lay between 50% and 100%. On the larger benchmarks it led the strongest competitor or was not statistically distinguishable from it; these cases, and the one loss (to an equal-data three-action program under very strong demand correlation, which the state-conditional variant reverses), occur where there is little to save or where the dependence bias of the fresh-start value is strongest. On four of the six planning gates it also exceeded the clairvoyant bound of the handoff-only class; that bound is clairvoyant only about the handoff lever, so the excess is a measure of what the second lever is worth, not a contradiction.

For the practitioner, five findings travel beyond our test bed. First, *label on the peak.* Aggregated per-route data invite risk labels computed from end-of-route totals, and Proposition 1 shows the resulting policies can be blind to the very event they are meant to prevent; monitoring must record the running maximum. Second, *the break-even threshold over-triggers by construction, and the premium can be bounded.* Proposition 3 bounds it by the cost of interventions on days that would have completed cleanly and of acting before cheaper later opportunities—under flat prices, by the handoff price times the probability of such clean-day interventions. A threshold tuned on realised cost moves away from the break-even level and is not covered by the bound; its excess cost over the handoff-only policy is measured separately in §4.4. On tightly packed plans and short routes that premium is small, and a tuned threshold is a defensible simplification of the handoff decision; on plans that carry slack it is larger, and there the threshold family saved little in our experiments while the priced policy still earned double-digit savings. Third, *planning conservatism and execution intelligence are substitutes, but asymmetric ones*: robust plans leave less recourse cost to

save, yet what remains is recovered best by the policy that prices the full action menu. Fourth, *plan slack and the demand profile select the lever.* Depot returns earn most on plans that carry slack and on geometries with central depots—classical benchmark geometries flatter them—and little on packed plans; on our delivery-dominated city instances, planned without slack, they were nearly worthless. A deployed policy must discover this per route, and on our city instances BATON does so and reduces to its handoff-only form. With the instances available we cannot separate the role of the road network from that of plan slack and demand profile, so we do not claim that urban geography as such makes the return worthless. Fifth, *buy standby capacity by the use unless it can be held cheaply.* A reserved standby vehicle serves at most one route a day, so at a holding cost equal to the standby day rate a reserved pool cannot beat pay-per-use; below that rate it pays where handoffs are frequent, on packed plans and above all on pools shared by many plans. Where capacity is capped, deciding each route with a shadow price on the pool beats ignoring the cap, keeps the routes decoupled, and relies on the depot return as the fallback lever (§4.6).

The structural results behind these findings do not rest on hand-checked algebra alone: Propositions 1–4 are verified in Lean 4, within the modelling abstractions listed at the start of §3.

Scope of the conclusions. Our managerial conclusions hold for the model we studied, and several features of real operations lie outside it. The model has no time windows: lateness is priced per downstream customer, uniformly, and customers with heterogeneous delay penalties or hard windows would make the handoff and return prices depend on which customers remain, not only on how many. The routes are fixed and executed in their planned order; we do not re-sequence the remaining stops, and we do not integrate execution with route planning. Shared resources enter only through the standby pool of §4.6, with a first-come-first-served dispatch rule and a shadow price chosen per pool; depot congestion, driver hours and standby vehicles that serve several

routes in one day are not modelled, and a standby vehicle is assumed to meet the route at the next stop without waiting, so its response time enters only through the transfer dwell. Demand is simulated from a calibrated protocol rather than drawn from operational logs, and the fleet costs are calibrated to orders of magnitude rather than to one operator's rate card. The comparisons between execution policies are, we believe, robust to these choices, since every policy faces the same model and our sensitivity analyses move each of them; the absolute savings are not, and should be read as indicative.

Limitations and next steps. The policy conditions on a scalar load statistic. Our negative result shows that richer statistics do not pay at a thousand training paths, and the robustness study shows that the scalar state loses little under strongly dependent demand, but fleets with vastly longer logs may cross that boundary, and the recursion accepts any regressor. The action set is general in form but has definite boundaries. Proposition 4 and the algorithm cover any action that ends the route at a price known at the current stop or resets the load to a known level at a known price; handoffs and depot returns are of these kinds. A *partial* handoff—transferring only the riskiest remaining customers to a standby vehicle—is not: it changes the remaining route itself, so the state must record which suffix of the route the planned vehicle still serves, the price schedule must be recomputed for every possible split, and the continuation value becomes a family of functions indexed by the split point. The backward induction extends to that augmented state with one regression per stop and split, at a cost quadratic rather than linear in the route length, and awaits a per-customer cost model to be worth instantiating. The demand model treats days as exchangeable. The experiment of §4.5, which mixes two known day types but keeps the days exchangeable, shows what such variation does to a deployed policy: a policy fitted on ordinary days and applied unchanged on promotion days keeps most, but not all, of its saving, while a policy fitted on a history that contains both day types does no worse than separate fits per type, which see fewer days each. The

remedy for known day types is therefore to keep them in the training history; separate fits pay only when each type has a long history of its own. Temporal drift and serial dependence between days lie outside our experiments; where the mix may drift, the policy should be re-fitted on a rolling window and monitored. Finally, our deepest structural finding cuts toward planning: on deterministically packed routes most breach risk sits at the first stop, where no execution policy can act, so the larger prize is a planner whose acceptance gate internalises the fitted execution cost itself—co-optimising the routes with the recourse policy that will execute them. We regard this as the most important direction for future work, because it attacks the risk that no execution policy can reach, and we did not attempt it here for two reasons. It changes the object of study—the plans would then depend on the execution policy, so a comparison of execution policies on common plans, which is the design that isolates the contribution of this paper, would no longer be possible—and it requires the fitted execution cost of every candidate route inside the planner's inner loop, which is a separate computational question. The machinery developed here is cheap enough to make that co-optimisation a concrete possibility.

Acknowledgements

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Data availability

All instance generators, result files, and code sufficient to reproduce every table and figure are available in a public repository at `https://github.com/vinhqdang/stochastic_vrp`. Machine-checked proofs of Propositions 1–4, written in Lean 4 with the Mathlib library, are in the same repository, in the subdirectory `BatonProofs` of the manuscript directory. The repository states

that produced each submitted version are tagged (`baton-r1`, `baton-r2`), so that the code can be checked against the version under review.

Appendix A. Supplementary material

Appendix A.1. Proof of Proposition 1

Proof of Proposition 1. The inclusion is immediate since $W_m \le \max_{k\le m} W_k$. If $W_m \le 0$ a.s. and $B > 0$ then $Y^{\text{end}} = 0$ a.s., so any regression of Y^{end} on any covariate is the zero function on the support of the data; a trigger $\hat{p}_k(W_k) > \tau$ with $\tau > 0$ never fires, and the policy is the reactive one. The reactive policy is feasible, so $\Delta \ge 0$. For the upper bound, every policy adapted to $(\mathcal{F}_k)$ pays C_{fail} on $\{T = 1\}$, where no decision epoch precedes the breach, at least $\min\{\omega_F, C_{\text{fail}}\} = \omega_F$ on $\{2 \le T \le m\}$, and at least 0 on $\{T = \infty\}$; hence $V^\star \ge C_{\text{fail}}\mathbb{P}(T{=}1) + \omega_F\,\mathbb{P}(2 \le T \le m)$, and subtracting from $V^{\text{react}} = C_{\text{fail}}\,\mathbb{P}(T \le m)$ gives the right-hand inequality. It holds with equality when handing off at $T-1$ is an admissible policy, which is the case exactly when $T-1$ is a stopping time. For the lower bound, the policy "hand off after stop 1 unless the route has already breached" is admissible and costs $C_{\text{fail}}\mathbb{P}(T{=}1) + \omega_F\,\mathbb{P}(T \ge 2)$, the event $\{T \ge 2\}$ including $\{T = \infty\}$; subtracting its cost from V^{react} bounds Δ from below. □

Appendix A.2. Supplementary tables and figure

The following tables and figure support statements of §4 and are referred to there: the structural synthetic scenarios (Table A.1), the reinforcement-learning comparison (Table A.2), the fresh-start value and the post-return state (Table A.3), the training-data budget (Table A.4) and the offline fitting times (Table A.5), and a full-plan replay on the Hanoi network (Figure A.1).

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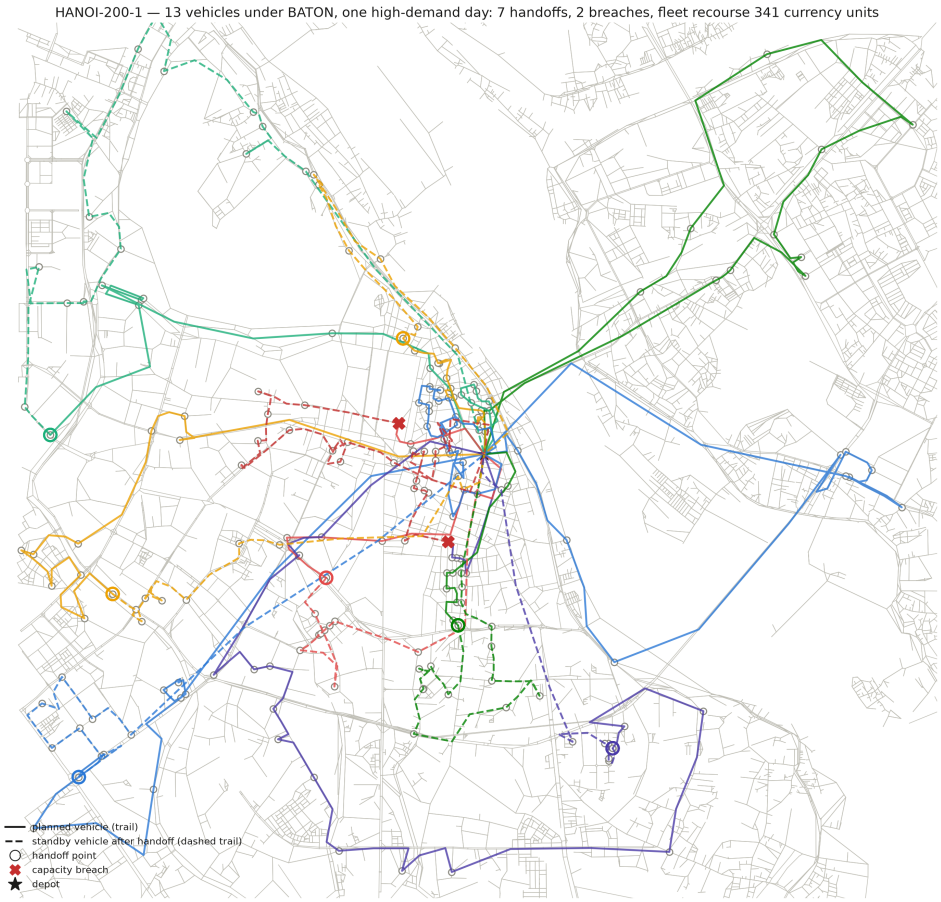


Figure A.1: One high-demand day for a full 200-customer plan on the Hanoi network (instance HANOI-200-1): thirteen vehicles execute simultaneously under BATON, sharing one demand scenario. Solid trails are planned vehicles, dashed trails the standby vehicles that take over at the circled handoff points, and red crosses the two spikes that outran any profitable anticipative action and fell through to the emergency guard. Seven of thirteen routes exercised the handoff lever; the whole day's recourse bill is 341 currency units.

Table A.1: Structural synthetic scenarios (five seeds, 1.2×10^4 test routes each, flat prices $\omega_F = 1$, C_{fail}/ω_F as shown or 5): saving % over the reactive policy, *higher is better*. The synthetic routes have no geometry, so the depot return of the full BATON is priced flat at half a handoff; the first three columns use the handoff lever only. On collect-then-deliver structure the endpoint-labelled predecessor never intervenes.

Scenario	endpoint + τ	peak + τ	BATON-HO	BATON
collect-then-deliver	0.0	49.2	51.9	**69.1**
milk run, regime switching	58.9	58.9	63.8	**78.5**
milk run, $C/\omega = 20$	85.5	85.5	88.2	**93.5**

Table A.2: Reinforcement-learning baseline (re-implementation of the policy architecture of Iklassov et al., 2024) versus BATON on the identical 49 routes and out-of-sample test days; saving % vs. the reactive policy, *higher is better*. The learned policy acts on the handoff lever only, so BATON-HO is the like-for-like comparison; the full BATON row adds the depot return.

Policy	saving vs. reactive (%)
RL, 40 epochs	27.2
RL, 150 epochs, seed 0	27.4
RL, 150 epochs, seed 1	27.3
RL, 150 epochs, seed 2	27.3
BATON-HO (handoff only)	**32.9**
BATON (handoff and depot return)	**33.7**

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Table A.3: The fresh-start value and the post-return state. Three quantities are compared at each stop: BATON's in-sample estimate $\widehat{F}_k$; the population value F_k^{pop} of the same fitted downstream policy from the reset state, estimated on 5×10^4 independent paths; and an approximate reference F_k^{ref} for the optimal value, the fresh-start value of a *constant-F* high-data plug-in program ($\mathrm{DP}^{3,c}_{50\mathrm{k}}$, the unconditional counterpart of $\mathrm{DP}^{3}_{50\mathrm{k}}$, fitted on 2.5×10^4 paths and valued on another 2.5×10^4). First two columns: $\widehat{F}_k - F_k^{\mathrm{pop}}$ (in-sample effect) and $F_k^{\mathrm{pop}} - F_k^{\mathrm{ref}}$ (downstream suboptimality plus reference error), mean over stops and routes in % of the handoff price, with 95% bootstrap intervals resampling instances. Saving columns (% vs. reactive, all *without* deployment selection, so that the three-action fit is exposed): the three-action policy as fitted, with $\widehat{F}_k$ replaced by F_k^{pop}, with F_k replaced by F_k^{ref}, with the state-conditional fresh-start value (BATON-CF), with the physical reset $x_k = -\min(D_{\le k}, L_0)$, and the state-conditional high-data program $\mathrm{DP}^{3}_{50\mathrm{k}}$ of the main tables. Savings in this table pool the route costs of all plans of a benchmark (ratio of sums), whereas Tables 2 and 5 average the plan-level savings over instances, so the two differ slightly.

	error of F (% of H) [95% CI]		three-action BATON, saving %						
Benchmark	$\widehat{F} - F^{\mathrm{pop}}$	$F^{\mathrm{pop}} - F^{\mathrm{ref}}$	BATON-HO	as fitted	F^{pop}	F^{ref}	CF	physical reset	$\mathrm{DP}^{3}_{50\mathrm{k}}$
Salhi–Nagy (Det)	+0.40 [+0.2, +0.6]	+0.09 [-0.5, +0.6]	41.6	52.6	52.6	52.2	53.5	58.2	53.9
Dethloff (SAA)	-0.01 [-0.05, +0.04]	+0.01 [+0.00, +0.02]	18.9	51.4	51.4	51.4	52.2	54.6	54.6
City, real shops (Det)	-0.13 [-0.5, +0.2]	-1.09 [-1.7, -0.5]	9.0	5.5	5.4	5.7	9.2	9.7	10.5

Table A.4: Training-data budget: saving % vs. reactive as a function of the number N of training days per route (same test days throughout). thr.: label-corrected tuned threshold; DP_N: plug-in dynamic program at equal data; BATON with deployment selection. The last row gives the high-data reference programs (handoff-only / three-action). The $N = 10^3$ row re-draws the training days of a smaller instance set (Section 4.7) and therefore differs from Tables 2 and 5 by up to 0.3 points.

	Dethloff, DET			Dethloff, SAA			City, DET		
N	thr.	DP_N	BATON	thr.	DP_N	BATON	thr.	DP_N	BATON
100	23.0	22.7	25.8	-2.4	-1.6	41.7	0.5	-0.1	3.9
250	24.0	23.8	26.9	7.4	-0.5	49.7	5.5	1.3	8.1
500	24.6	25.3	27.6	13.0	2.6	52.0	8.5	3.4	9.5
1,000	24.8	25.7	27.9	16.5	12.1	53.6	9.9	7.0	10.4
2,000	25.0	25.7	28.0	18.1	18.3	54.8	10.8	9.3	11.5
5,000	25.0	25.8	28.2	18.9	21.1	55.6	11.5	10.4	12.1
20,000	25.0	26.3	28.3	19.6	22.0	56.3	12.0	11.9	12.9
ref.: $\mathrm{DP}_{50\mathrm{k}}$ / $\mathrm{DP}^{3}_{50\mathrm{k}}$		26.4	29.5		22.2	56.4		12.2	12.4

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Table A.5: Offline fitting time per route (single CPU core, milliseconds; median and 90th percentile over 102 routes of ten Dethloff SAA plans and four city plans). Every policy is fitted on the same 10^3 training days except the two reference programs. Tuning by grid search on realised training cost is included; the position-dependent threshold includes the fits of its two starting points. The online decision of every cost-comparison policy is one lookup in a monotone piecewise-linear function and two comparisons: 1.2 μs per decision through the fitted breakpoints (63 μs through the scikit-learn prediction call).

Policy	training days	median	p90
tuned threshold (peak label)	10^3	63.9	100.4
position-dependent threshold	10^3	155.7	343.7
π_3 rule (grid-tuned)	10^3	4.9	7.6
rollout	10^3	6.1	10.7
cost-scaled rollout	10^3	47.9	81.6
depot-return restocking	10^3	4.8	7.4
two-lever rule (joint grid)	10^3	413.4	677.6
DP_N (equal data)	10^3	3.1	5.0
DP^3_N (equal data)	10^3	7.0	14.4
BATON-HO	10^3	7.2	11.3
BATON (incl. deployment selection)	10^3	24.2	49.2
BATON-CF	10^3	25.0	58.6
$\mathrm{DP}_{50\mathrm{k}}$ (reference, 5×10^4 paths)	5×10^4	110.0	170.7
$\mathrm{DP}^3_{50\mathrm{k}}$ (reference, 5×10^4 paths)	5×10^4	279.8	573.8

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